



Data Visualization



SUMMARIZING DATA

➡ Tables

- ➡ Simplest way to summarize data
- ➡ Data is presented as absolute numbers or percentages

➡ Charts and graphs

- ➡ Visual representation of data
 - ➡ Usually data is presented using percentages
- 



Data Visualization

- Graphs and charts are effective visual tools because they present information quickly and easily. It is not surprising then, that graphs are commonly used by print and electronic media. Sometimes, data can be better understood when presented by a graph than by a table because the graph can reveal a **trend or comparison**.
 - Students also find that graphs are easy to use because graphs are made up of lines, dots and blocks—all geometric forms that are simple and quick for students to draw. In the world of statistics, graphs display the relationship between variables or show the value spread of a given variable or phenomenon.
- 



Using graphs

- Knowing how to convey information graphically is important in presenting statistics. The following is a list of general rules to keep in mind when preparing graphs.
 - A good graph:
 - accurately shows the facts,
 - grabs the reader's attention,
 - complements or demonstrates arguments presented in the text,
 - has a title and labels,
 - Is simple and uncluttered,
 - shows data without altering the message of the data,
 - clearly shows any trends or differences in the data,
 - is visually accurate (i.e. if one chart value is 15 and another 30, then 30 should appear to be twice the size of 15).
- 



Types of Graphs

➤ There are many different types of graphs that can be used to convey information, including:

- Histograms
- Bar charts
- Scatterplots
- Pie charts
- Line charts
- Ogive
- Dot Plot
- Steam and Leave

Histogram

- The histogram is a popular graphing tool. It is used to summarize discrete or continuous data that are measured on an interval scale. It is often used to illustrate the major features of the distribution of the data in a convenient form. It is also useful when dealing with large data sets (greater than 100 observations). It can help detect any unusual observations (outliers) or any gaps in the data.
- A histogram divides up the range of possible values in a data set into classes or groups. For each group, a rectangle is constructed with a base length equal to the range of values in that specific group and a length equal to the number of observations falling into that group. A histogram has an appearance similar to a vertical bar chart, but there are no gaps between the bars. Generally, a histogram will have bars of equal width. Chart 5.7.1 is an example of a histogram that shows the distribution of salary, a continuous variable, of the employees of a corporation.

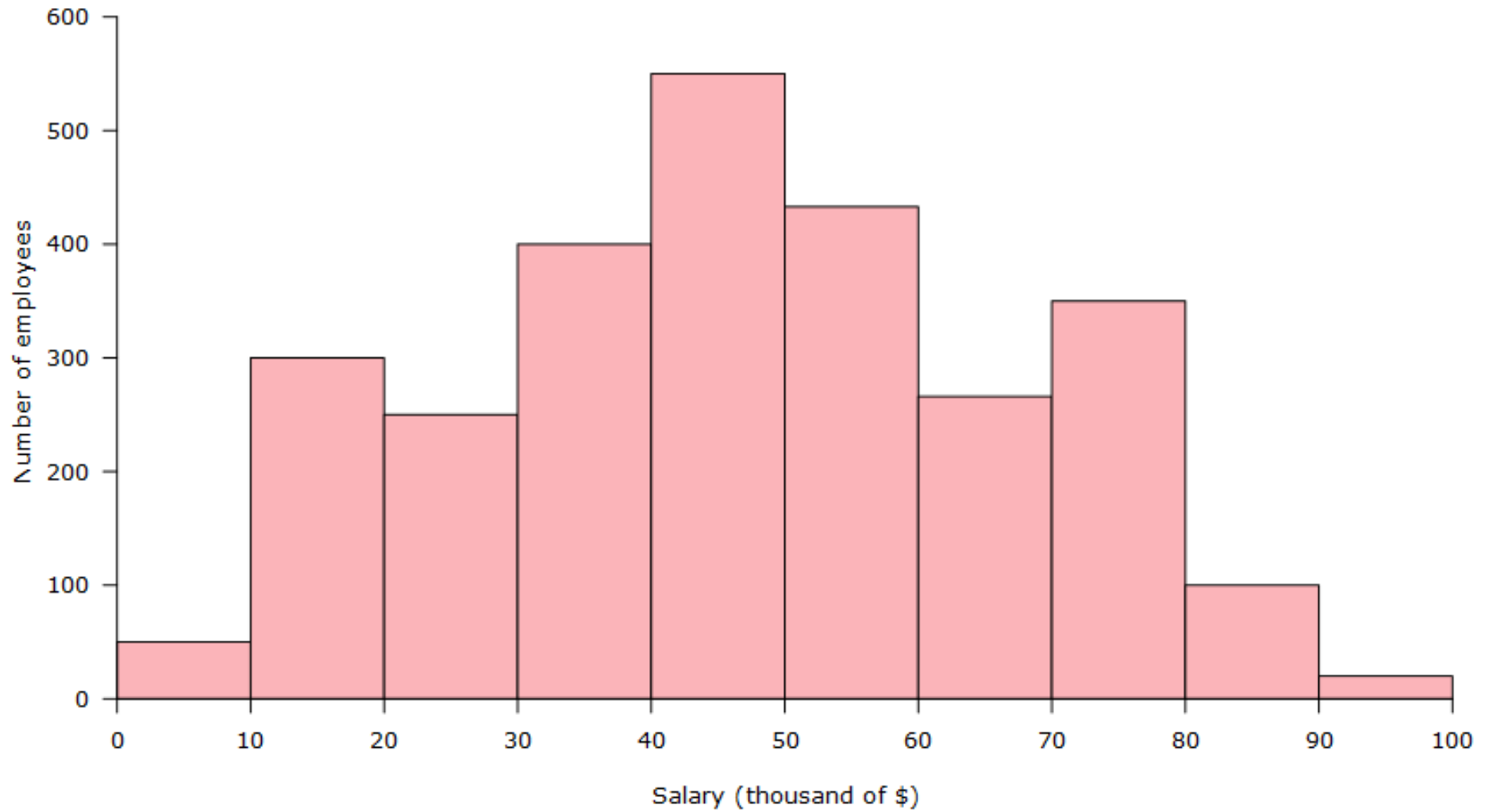


Data Table of the distribution of the salaries of employees of ABC corporation



Salary (in thousands of \$)	Number of employees
0–10	50
11–20	300
21–30	250
31–40	400
41–50	550
51–60	433
61–70	266
71–80	350
81–90	100
91+	20

Distribution of salaries of the employees of ABC Corporation



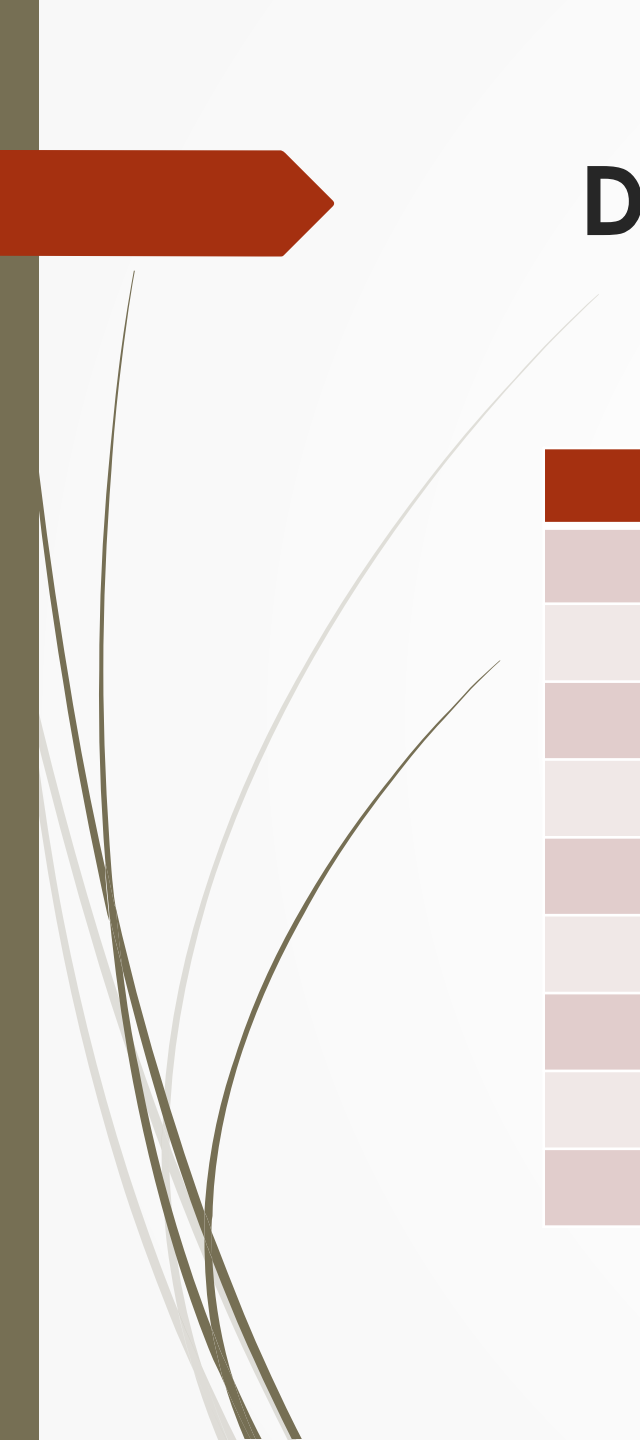


Bar chart

- ▶ A bar chart may be either horizontal or vertical. The important point to note about bar charts is their bar length or height—the greater their length or height, the greater their value. Bar charts are one of the many techniques used to present data in a visual form so that the reader may readily recognize patterns or trends.
 - ▶ Bar charts usually present categorical variables, discrete variables or continuous variables grouped in class intervals. They consist of an axis and a series of labelled horizontal or vertical bars. The bars depict frequencies of different values of a variable or simply the different values themselves. The numbers on the y-axis of a vertical bar chart or the x-axis of a horizontal bar chart are called the scale.
- 

Vertical bar charts

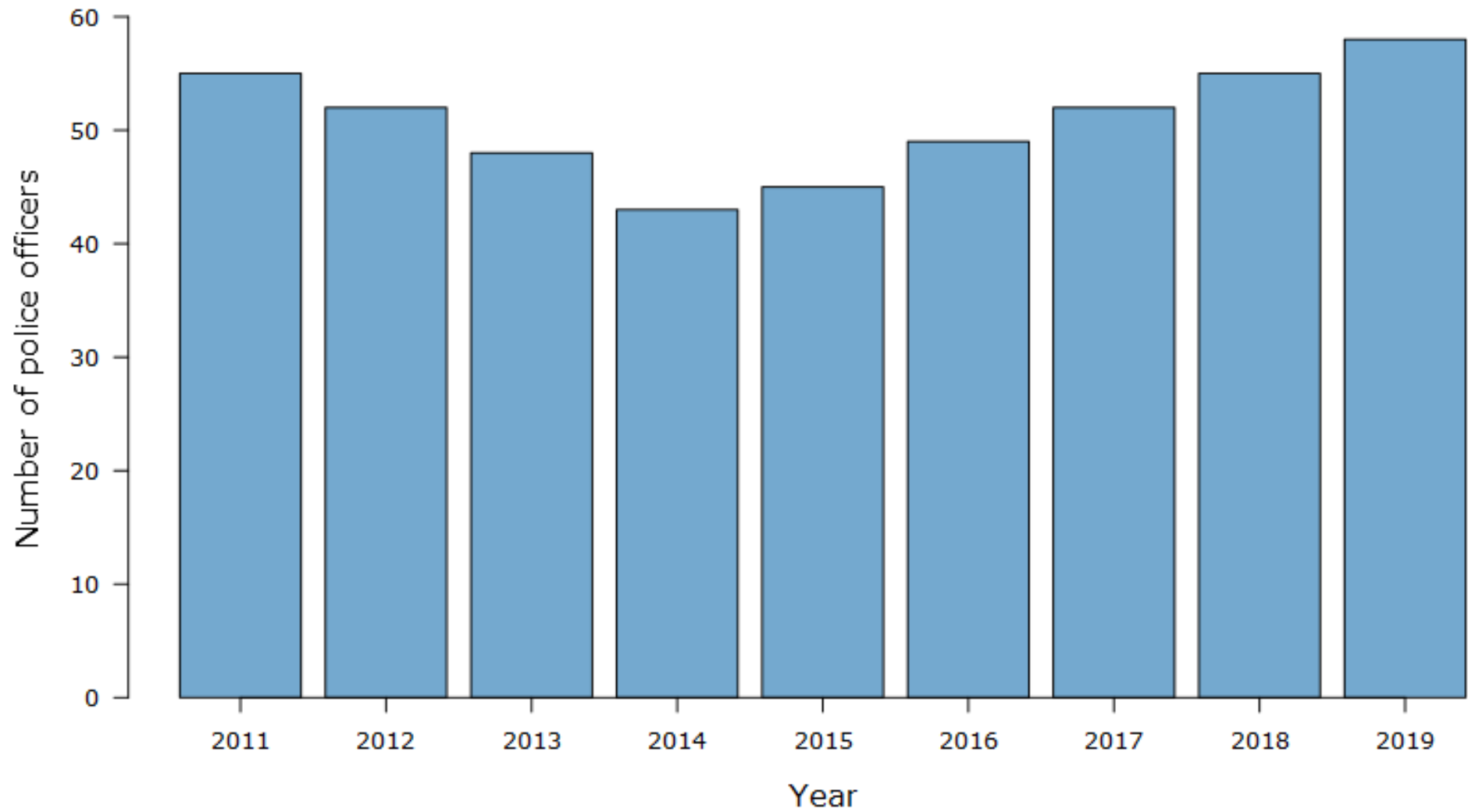
- Bar charts should be used when you are showing segments of information. Vertical bar charts are useful to compare different categorical or discrete variables, such as age groups, classes, schools, etc., as long as there are not too many categories to compare. They are also very useful for time series data. The space for labels on the x-axis is small, but ideal for years, minutes, hours or months. For example, Chart 5.2.1 below shows the number of police officers in Crimeville for each year from 2011 to 2019.



Data Table of number of police officers in crime Ville

Year	Number of police officers
2011	55
2012	52
2013	48
2014	43
2015	45
2016	49
2017	52
2018	55
2019	58

Number of police officers in Crimeville, 2011 to 2019





Grouped bar charts

- The grouped bar chart is another effective means of comparing sets of data about the same places or items. It gives two or more pieces of information for each item on the x-axis instead of just one as in Chart 5.2.1. This allows you to make direct comparisons on the same chart by age group, gender or anything else you wish to compare. However, if a grouped bar chart has too many series of data, the chart becomes cluttered and it can be confusing to read.

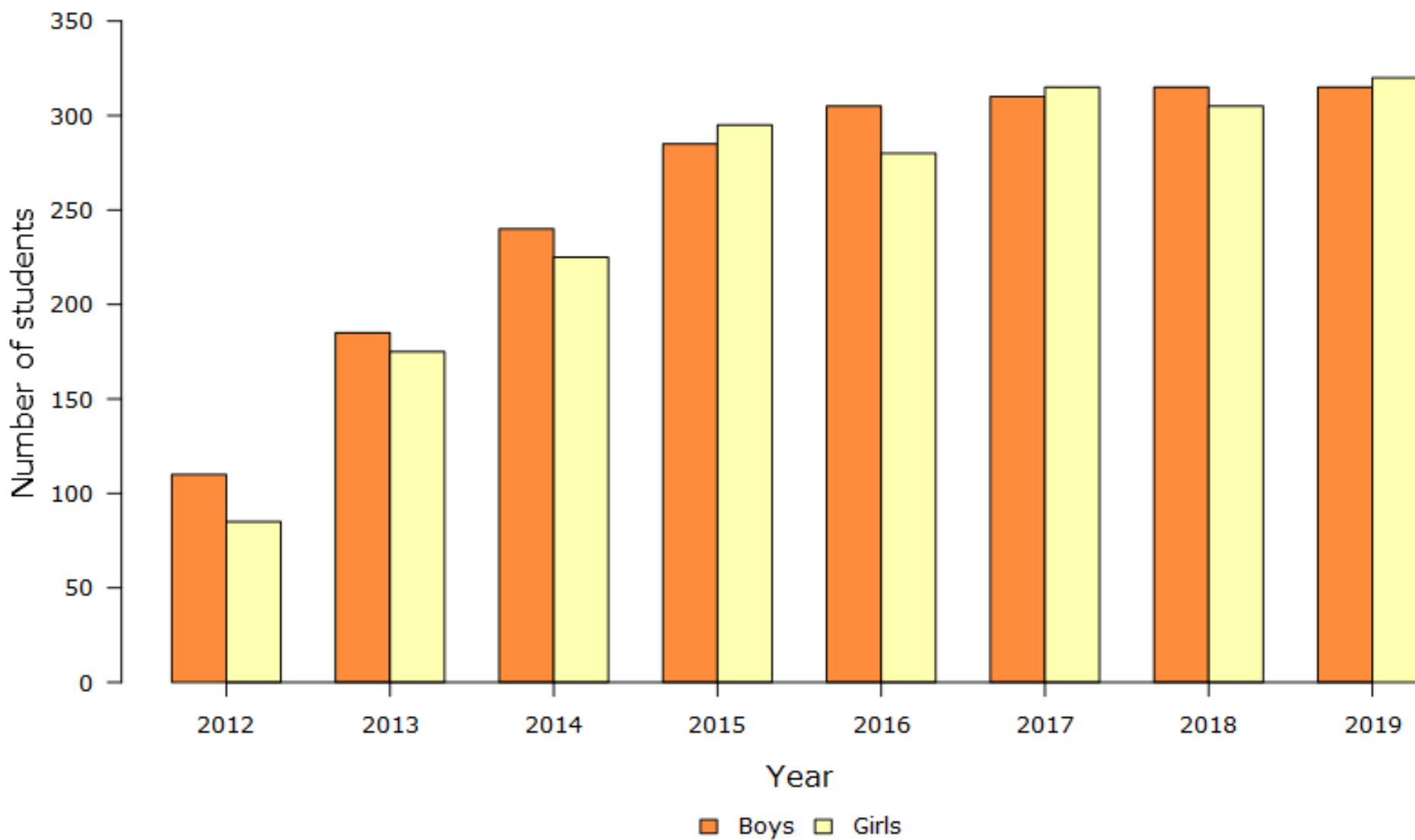


Data Table of the students who own a smartphone



Year	Number of boys	Number of girls
2012	110	85
2013	185	175
2014	240	225
2015	285	295
2016	305	280
2017	310	315
2018	315	305
2019	315	320

Students who own a smartphone at Redwood School, by gender, 2012 to 2019



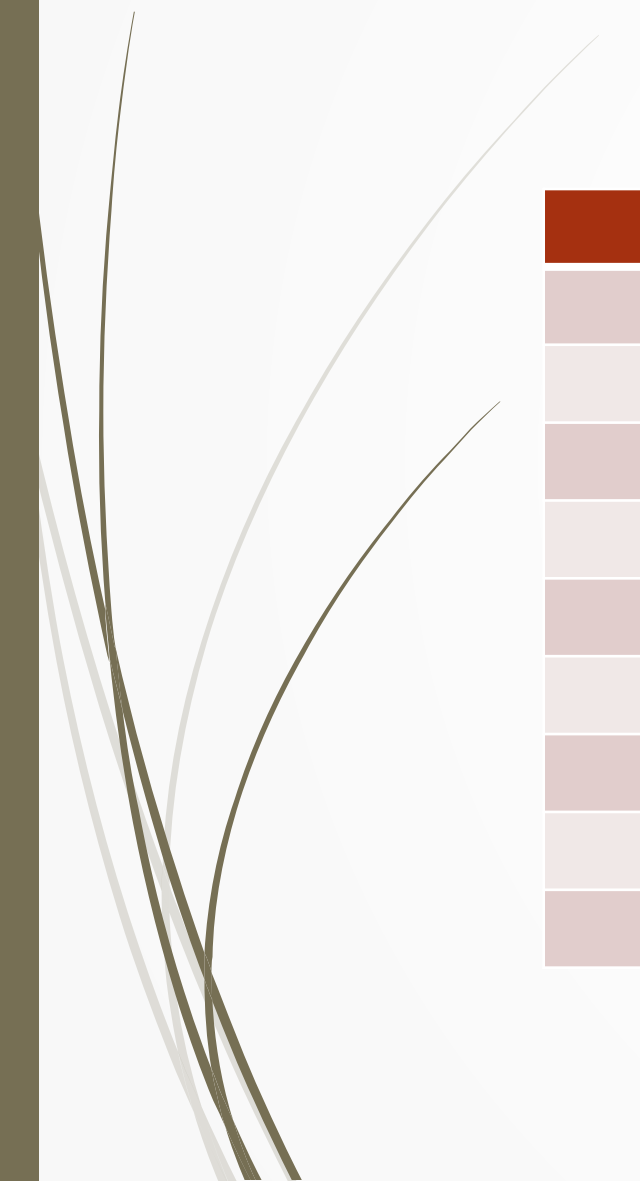


Horizontal bar charts

- One disadvantage of vertical bar charts, however, is that they lack space for text labelling at the foot of each bar. When category labels in the chart are too long, you might find a horizontal bar chart better for displaying information,

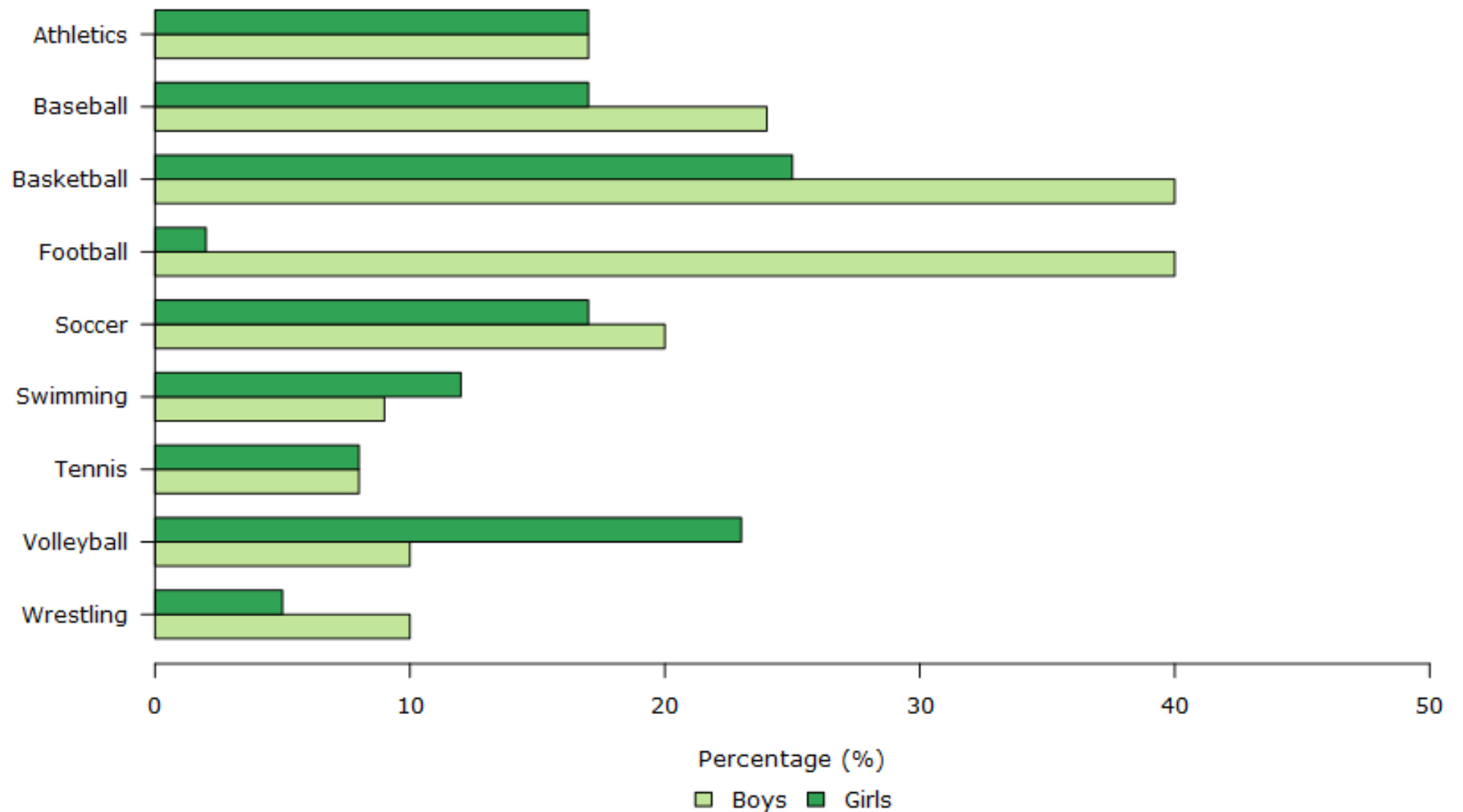


Data Table of the students who participated by 15 year.



Sport	Percentage of boys (%)	Percentage of girls (%)
Athletics	17	17
Baseball	24	17
Basketball	40	25
Football	40	2
Soccer	20	17
Swimming	9	12
Tennis	8	8
Volleyball	10	23
Wrestling	10	5

Sports practiced by 15-year-old students in Jamie's school, by gender



Stacked bar charts

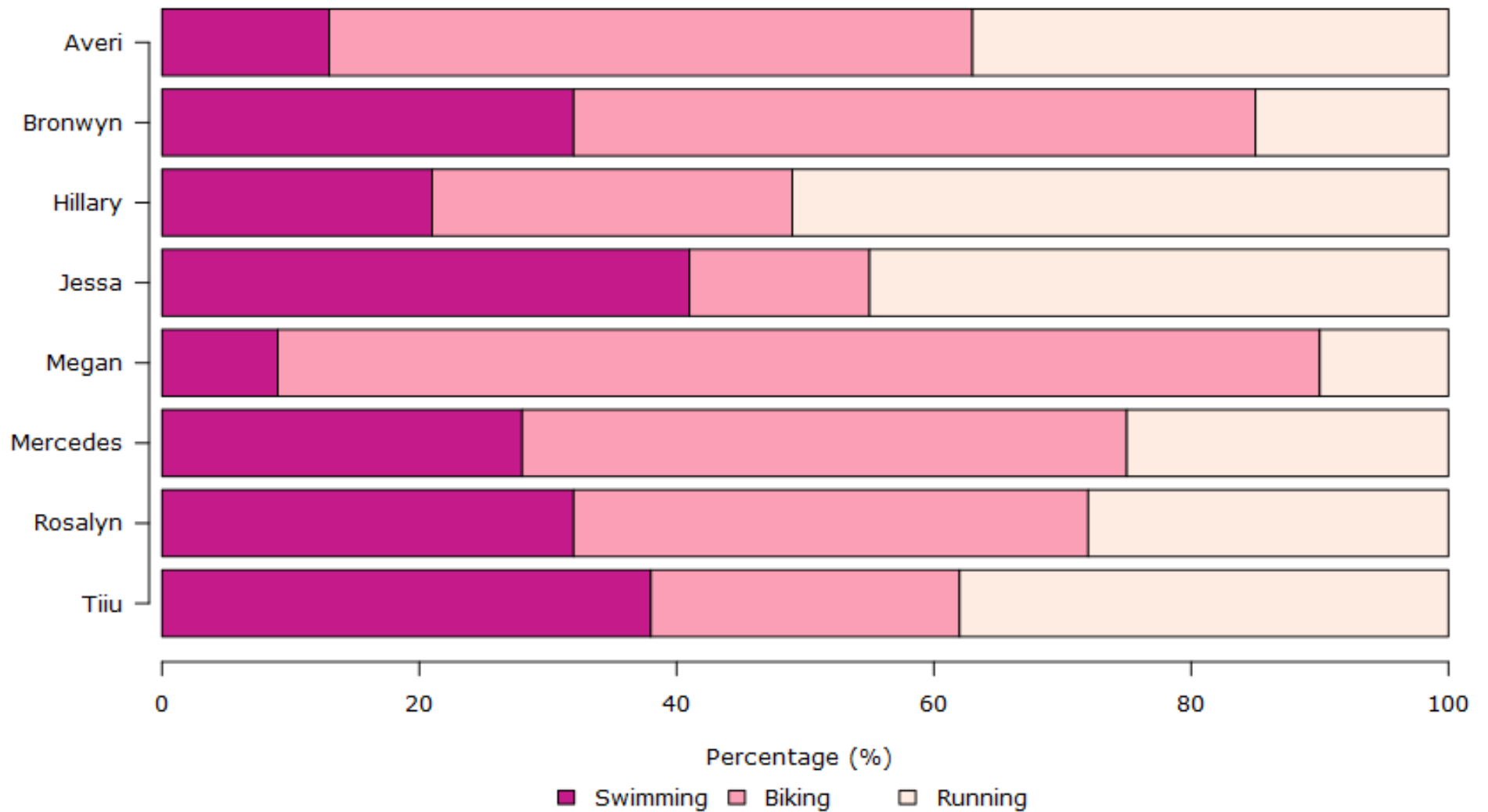
- There are several other types of bar chart that you may encounter. The population pyramid is a special application of a grouped bar chart. Another useful type of bar chart is the stacked bar chart.
- The stacked bar chart is a preliminary data analysis tool used to show segments of totals. The stacked bar chart can be very difficult to analyze if too many items are in each stack. It can contrast values, but not necessarily in the simplest manner.



Data Table of the percentage of time spent on each event

First name	Percentage of time spent swimming (%)	Percentage of time spent cycling (%)	Percentage of time spent running (%)
Averi	13	50	37
Bronwyn	32	53	15
Hillary	21	28	51
Jessa	41	14	45
Megan	9	81	10
Mercedes	28	47	25
Rosalyn	32	40	28
Tiiu	38	24	38

Campbell High Triathlon, percentage of time spent on each event, by competitor





Pie chart

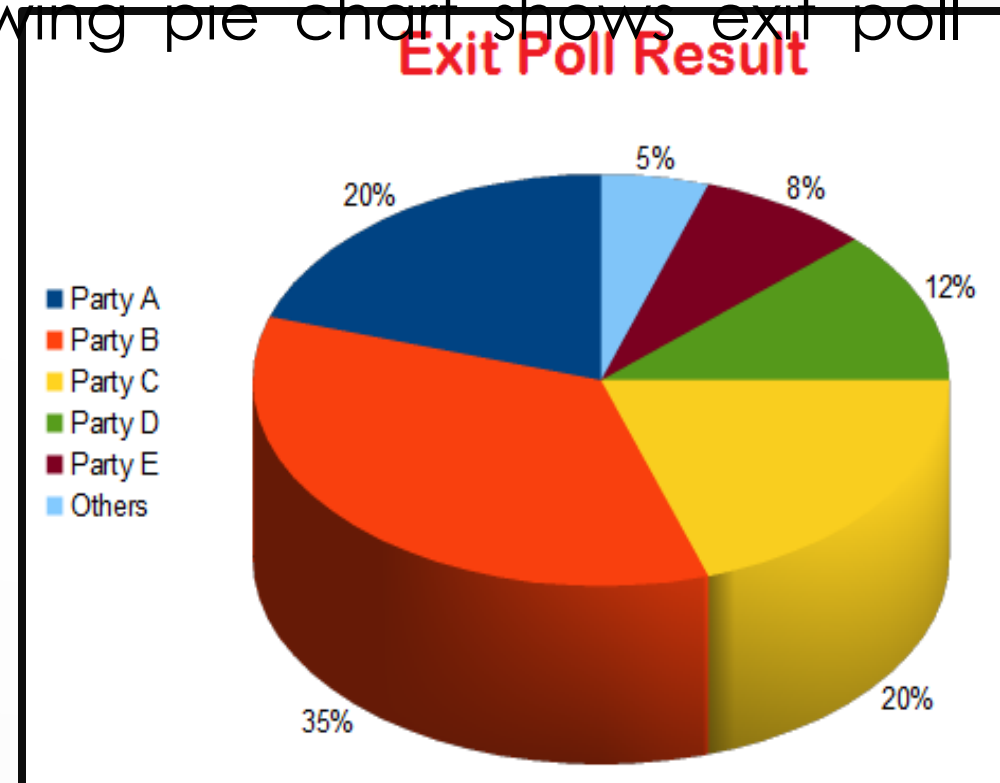
- 
- A pie chart, sometimes called a circle chart, is a way of summarizing a set of nominal data or displaying the different values of a given variable (e.g. percentage distribution). This type of chart is a circle divided into a series of segments. Each segment represents a particular category. The area of each segment is the same proportion of a circle as the category is of the total data set.

Pie chart

Pie chart is named so, due to its **similarity with pie and its slices**. Pie chart is a **circular chart** which is **divided into sectors**. **Each sector corresponds to different values of a data**. Generally, in pie charts, the percentage of each number is calculated and then obtained percentages are plotted at pie chart. Following pie chart shows exit poll results in an election.

Percentage of a data item can be calculated by the following formula:

$$\text{Percent Value} = \frac{\text{Data item} \times 100}{\text{Total}}$$



Cont...

Example: Imagine you survey with your friends to find which kind of movies they like best:

Table: Favorite Type of Movie				
Comedy	Action	Romance	Drama	SciFi
4	5	6	1	4

Solution:

Step 1: First, put your data into a table (like above), then add up all the values to get a total:

Table: Favorite Type of Movie					
Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20

Cont...

Step 2: Next, divide each value by the total and multiply by 100 to get a percent:

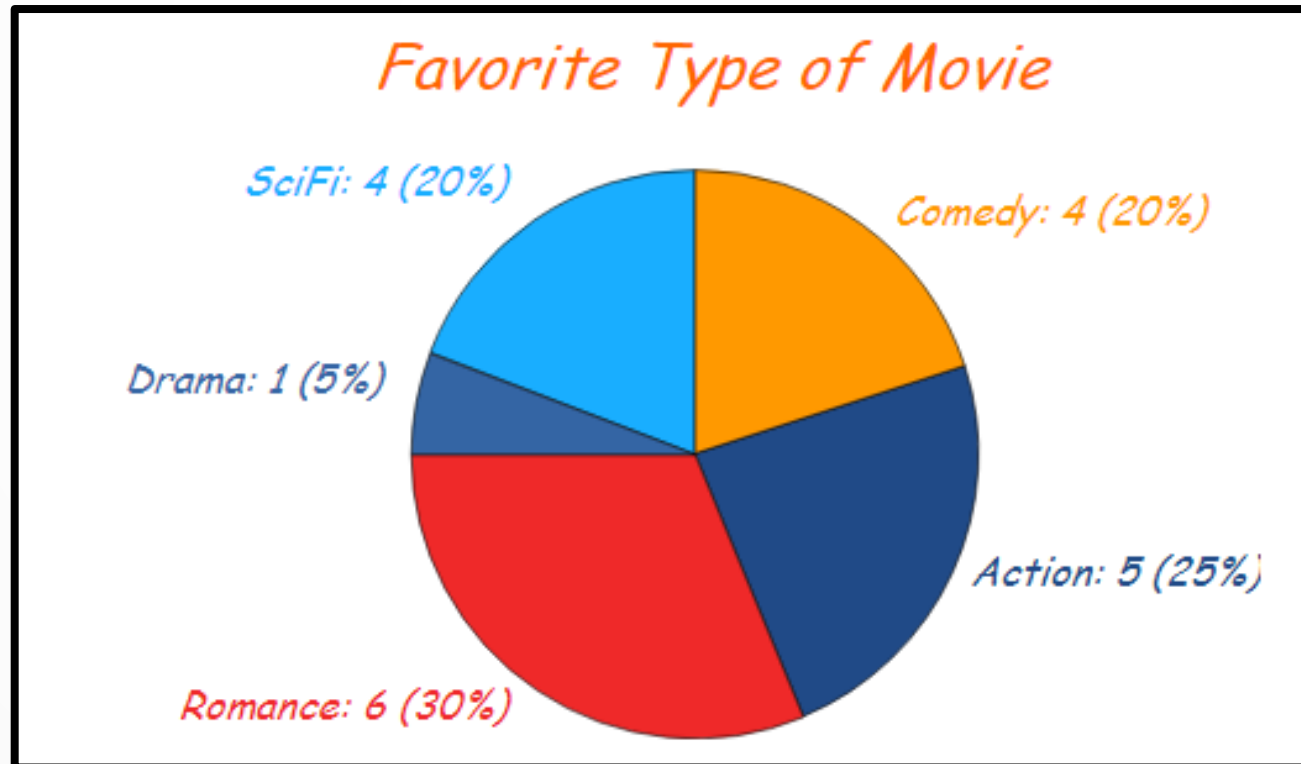
Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20
$4/20$ = 20%	$5/20$ = 25%	$6/20$ = 30%	$1/20$ = 5%	$4/20$ = 20%	100%

Step 3: Now to figure out how many degrees for each "pie slice" (correctly called a sector). A Full Circle has **360 degrees**, so we do this calculation:

Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20
20%	25%	30%	5%	20%	100%
$4/20 \times 360^\circ$ = 72°	$5/20 \times 360^\circ$ = 90°	$6/20 \times 360^\circ$ = 108°	$1/20 \times 360^\circ$ = 18°	$4/20 \times 360^\circ$ = 72°	360°

Cont...

Step 4: Then use your protractor to measure the degrees of each sector. Finish up by coloring each sector and giving it a label like "**Comedy: 4 (20%)**", etc.



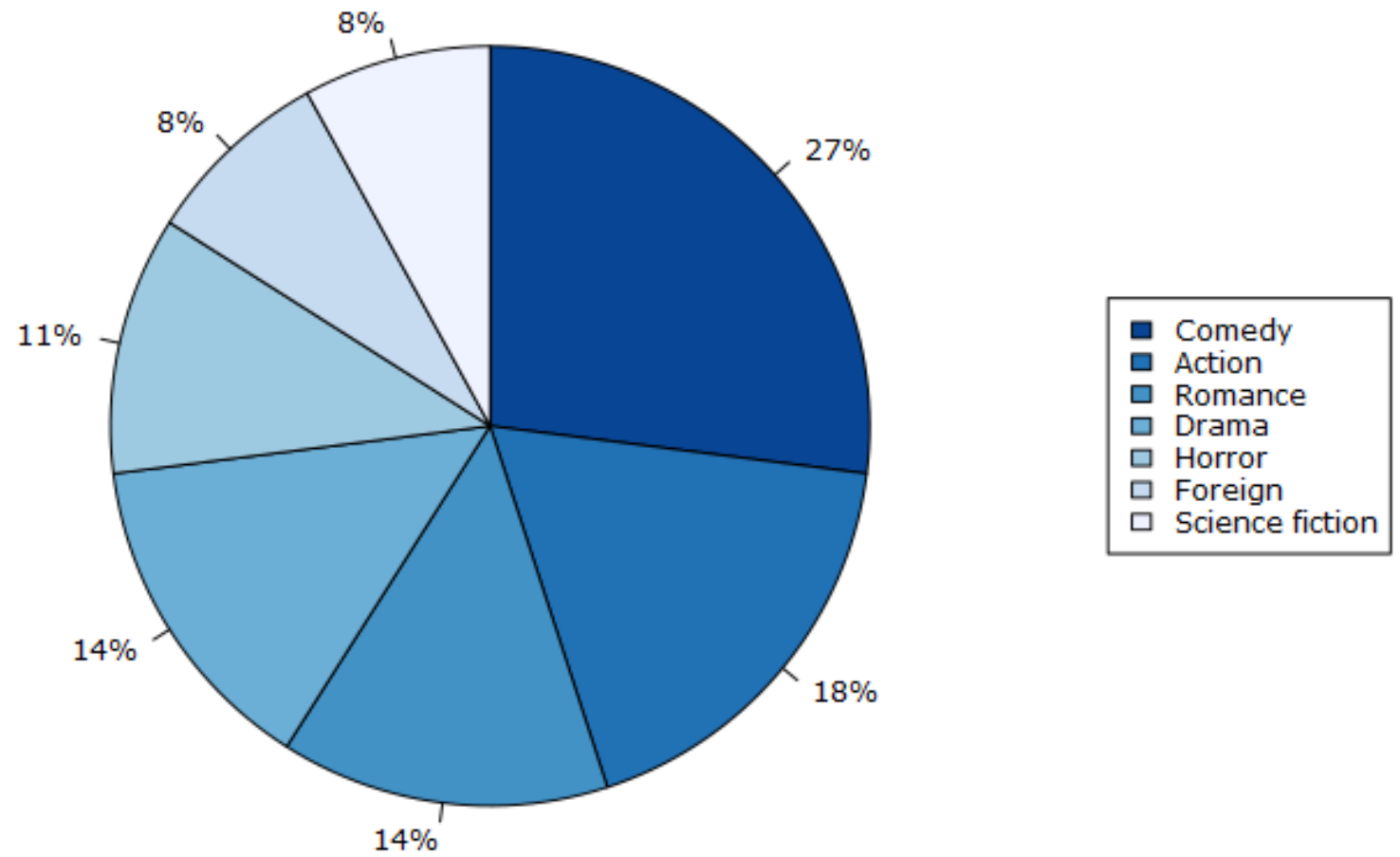


Data Table of the favorite movie of students



Movie genres	Percentage of students (%)
Comedy	27
Action	18
Romance	14
Drama	14
Horror	11
Foreign	8
Science fiction	8

Favourite movie genres in Mrs Smyth's Film class



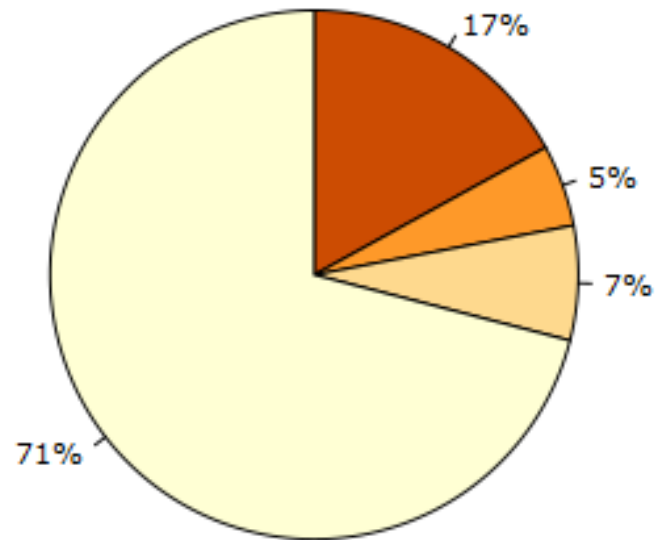


Data Table of the student who smoke by 15 - year

Smoking frequency	Percentage of boys (%)	Percentage of girls (%)
Every day	17	21
At least once a week	5	5
Less than once a week	7	6
Never	71	68

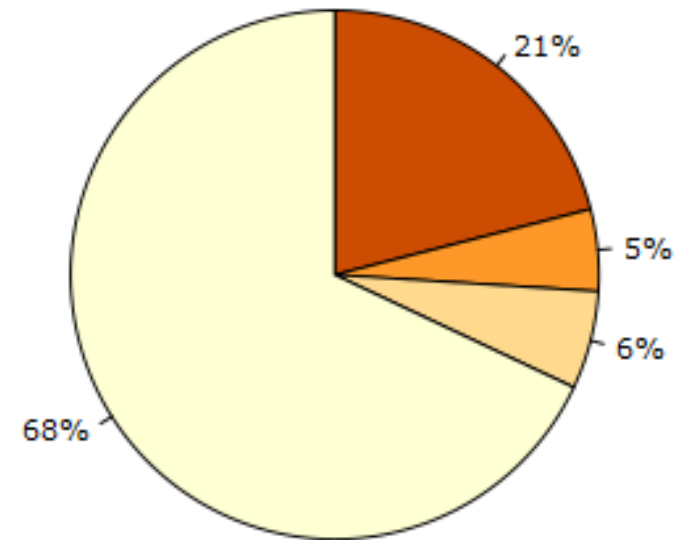
Smoking frequency of 15-year-olds on the Parkview Secondary School track and field team, by gender

Boys



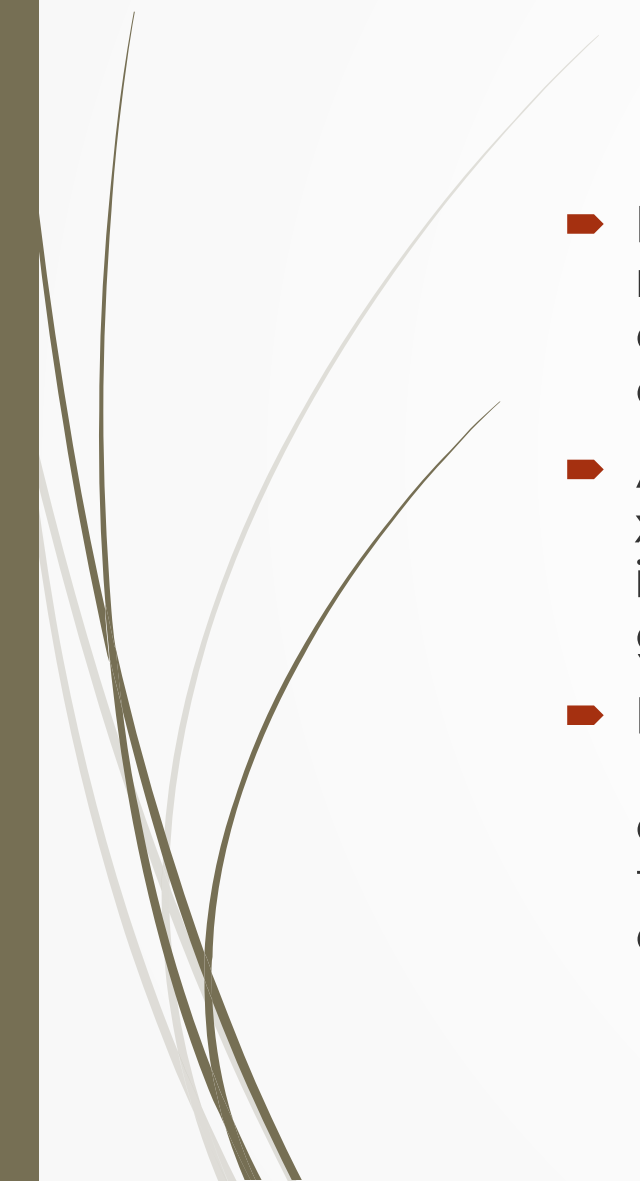
- Every day
- At least once a week
- Less than once a week
- Never

Girls





Line chart

- 
- ▶ Line charts, especially useful in the fields of statistics and science, are more popular than all other graphs combined because their visual characteristics reveal data trends clearly and these charts are easy to create.
 - ▶ A line chart is a visual comparison of how two variables—shown on the x- and y-axes—are related or vary with each other. It shows related information by drawing a continuous line between all the points on a grid.
 - ▶ Line charts compare two variables: one is plotted along the x-axis (horizontal) and the other along the y-axis (vertical). The y-axis in a line chart usually indicates quantity (e.g. dollars, litres) or percentage, while the horizontal x-axis often measures units of time. As a result, the line chart is often viewed as a time series graph.

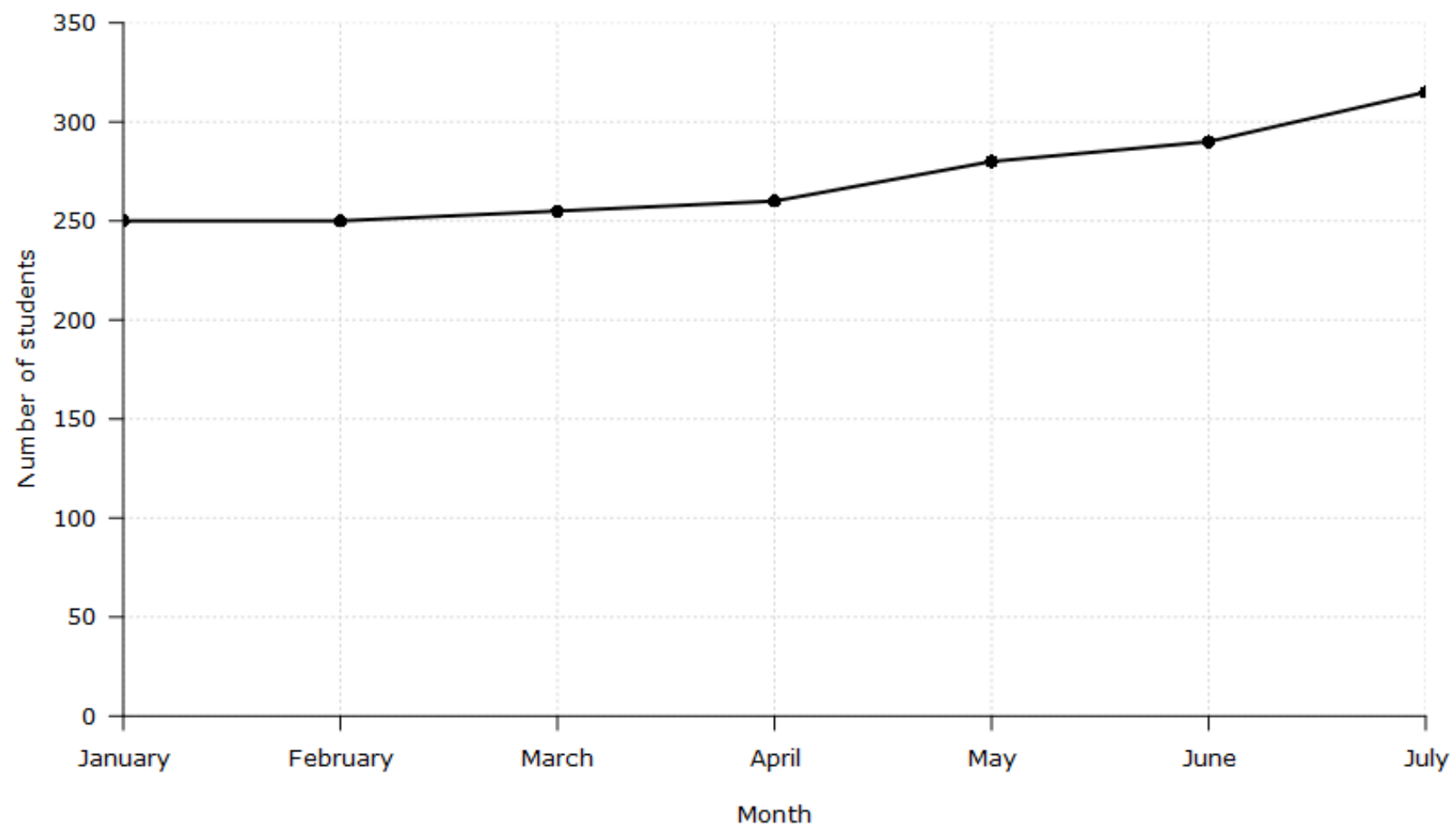


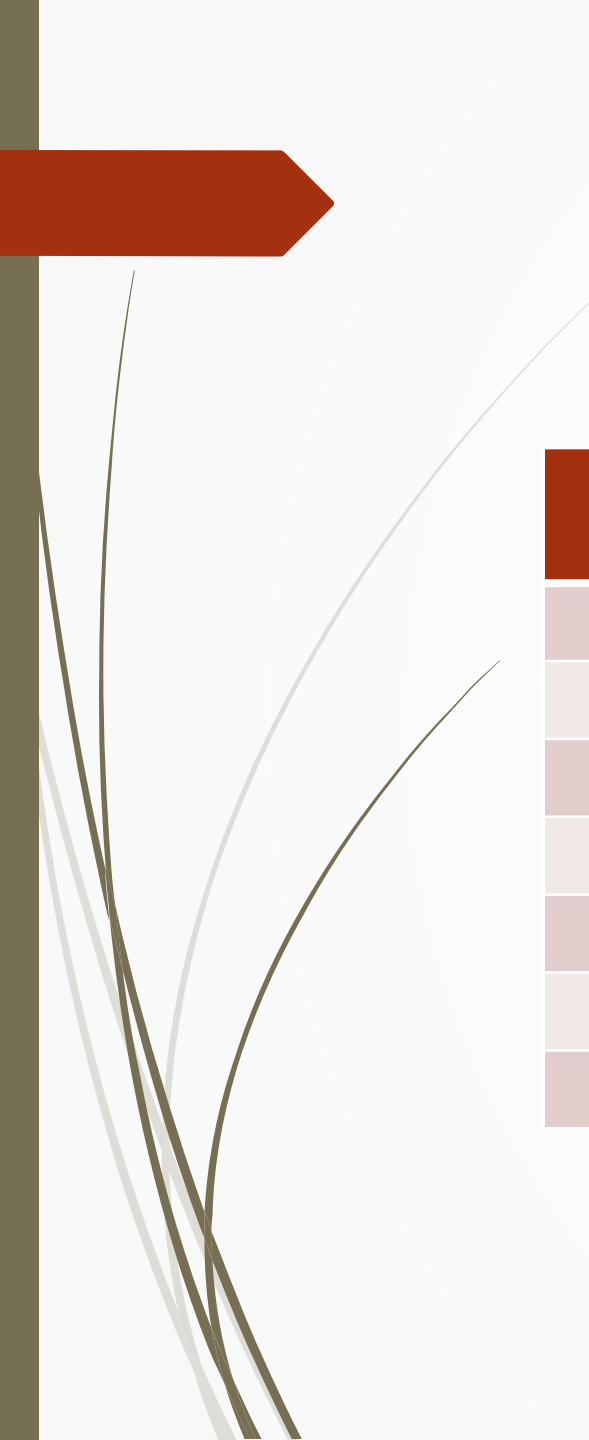
Data Table of students participating in the labour force

The number of students participating in the labour force was 252 in January, 252 in February, 255 in March, 256 in April, 282 in May, 290 in June and 319 in July. When examined further, the line chart indicates that the labour force participation of these students was at a plateau for the first four months (January to April), and for the next three months (May to July) the number increased steadily.

Month	Number of students
January	250
February	250
March	255
April	260
May	280
June	290
July	315

Labour force participation in Andrew's high school

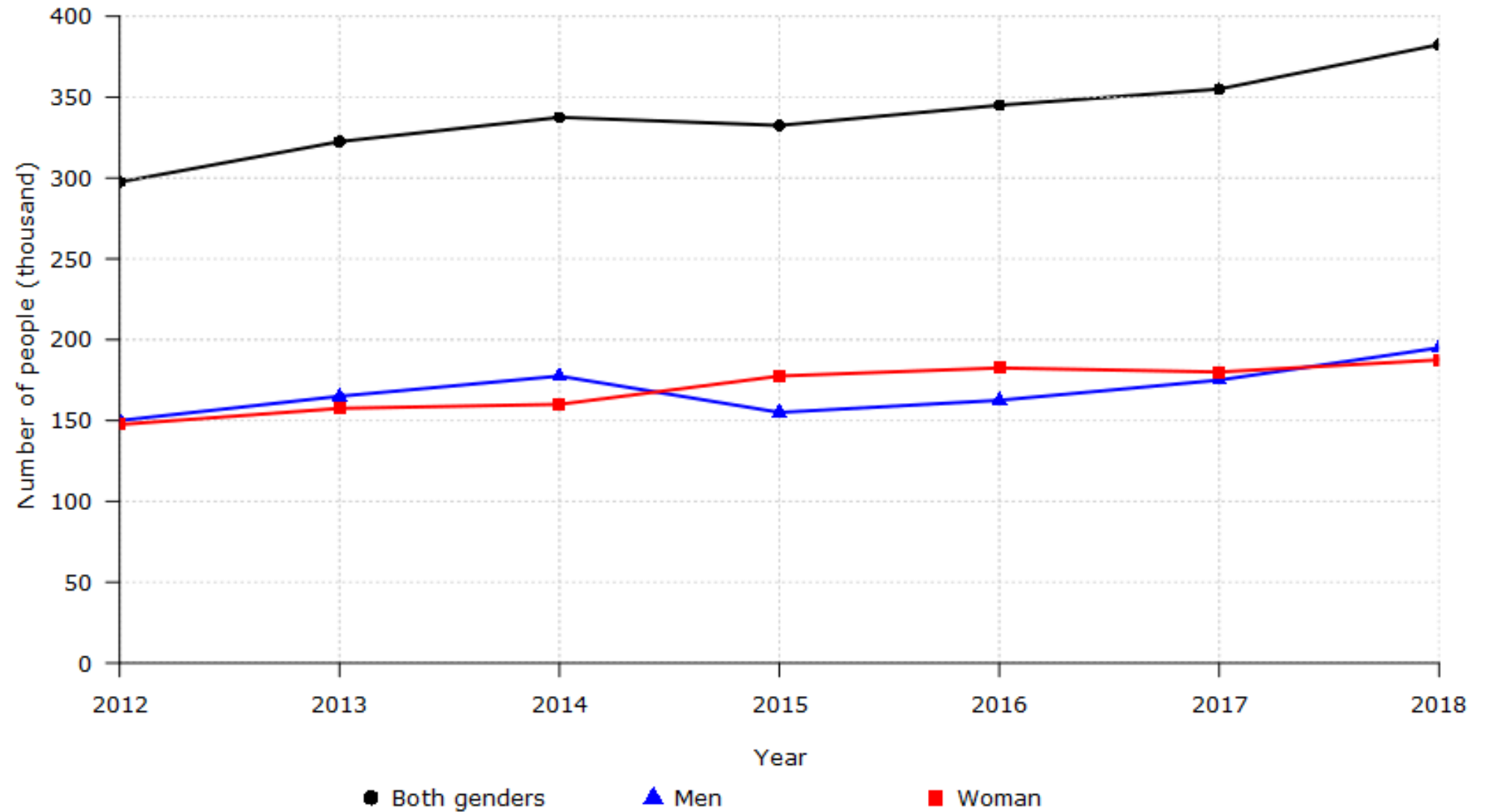




Data Table of smartphone uses in any Town

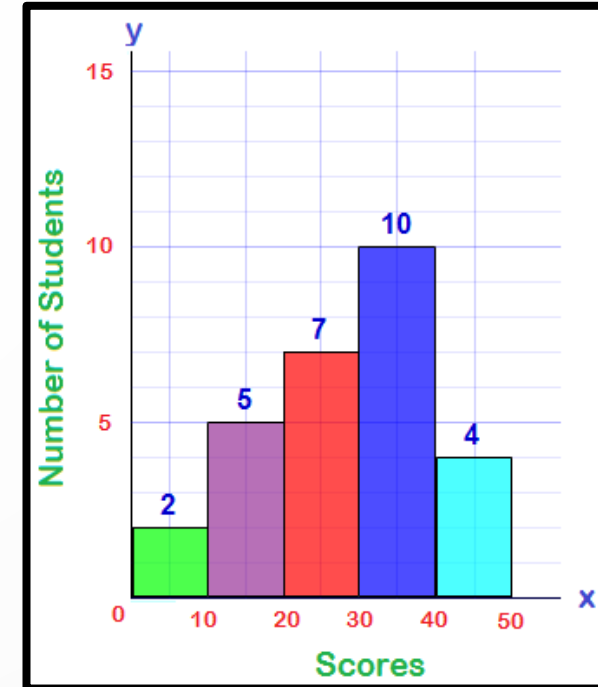
Year	Number of men (thousands)	Number of women (thousands)	Total number (thousands)
2012	150.0	147.5	297.5
2013	165.0	157.5	322.5
2014	177.5	160.0	337.5
2015	155.0	177.5	332.5
2016	162.5	182.5	345.0
2017	175.0	180.0	355.0
2018	195.0	187.5	382.5

Cell phone use in Anytown by gender, 2012 to 2018



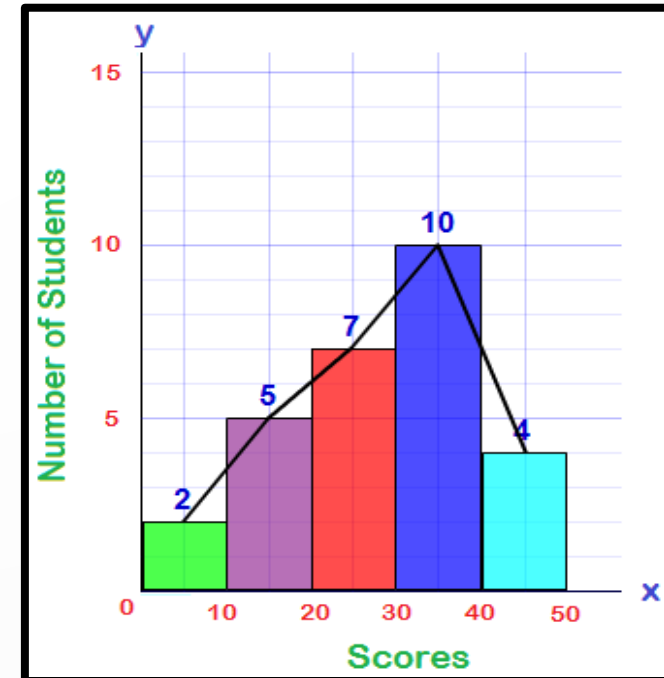
Histogram

Histogram is a graphical representation which **uses rectangular bars** to present the data. Histograms look like bar graphs. Unlike bar graphs, histograms **do not have gaps between two consecutive bars**. Histograms are most useful for representing grouped data. **Width of rectangular bar on X-axis usually denotes class interval.** Following graph demonstrates a histogram:



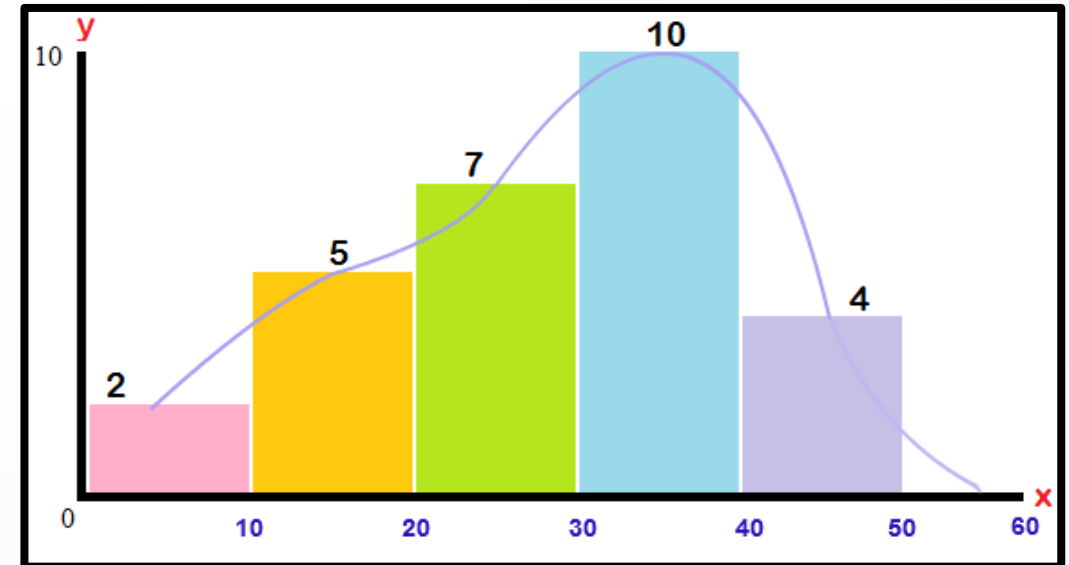
Frequency polygon

Frequency polygon is an **extension of histogram**. Frequency polygon is constructed by **marking mid points of all the bars in histogram** and **then, joining them together by straight lines**. Though, frequency polygon can be made without drawing histogram by just marking midpoints (of given class interval) on X-axis and then plotting points corresponding to them, join these points with straight lines.



Frequency Curve

A curve is a line that is **smooth and continuous**. In a histogram, **when midpoints are joined together with straight lines, a frequency polygon is formed**. **On the other hand, when midpoints are joined together by free hand resulting in a smooth curve, a frequency curve is formed**. It means that the **only difference** between frequency polygon and frequency curve is that unlike frequency polygon, frequency curve **is smooth and without sharp edges**. A diagram showing frequency curve of the same data used before is given below:



Distribution of a Data Set

The distribution of a data set is a table, graph, or formula that provides the values of the observations and how often they occur.

Example

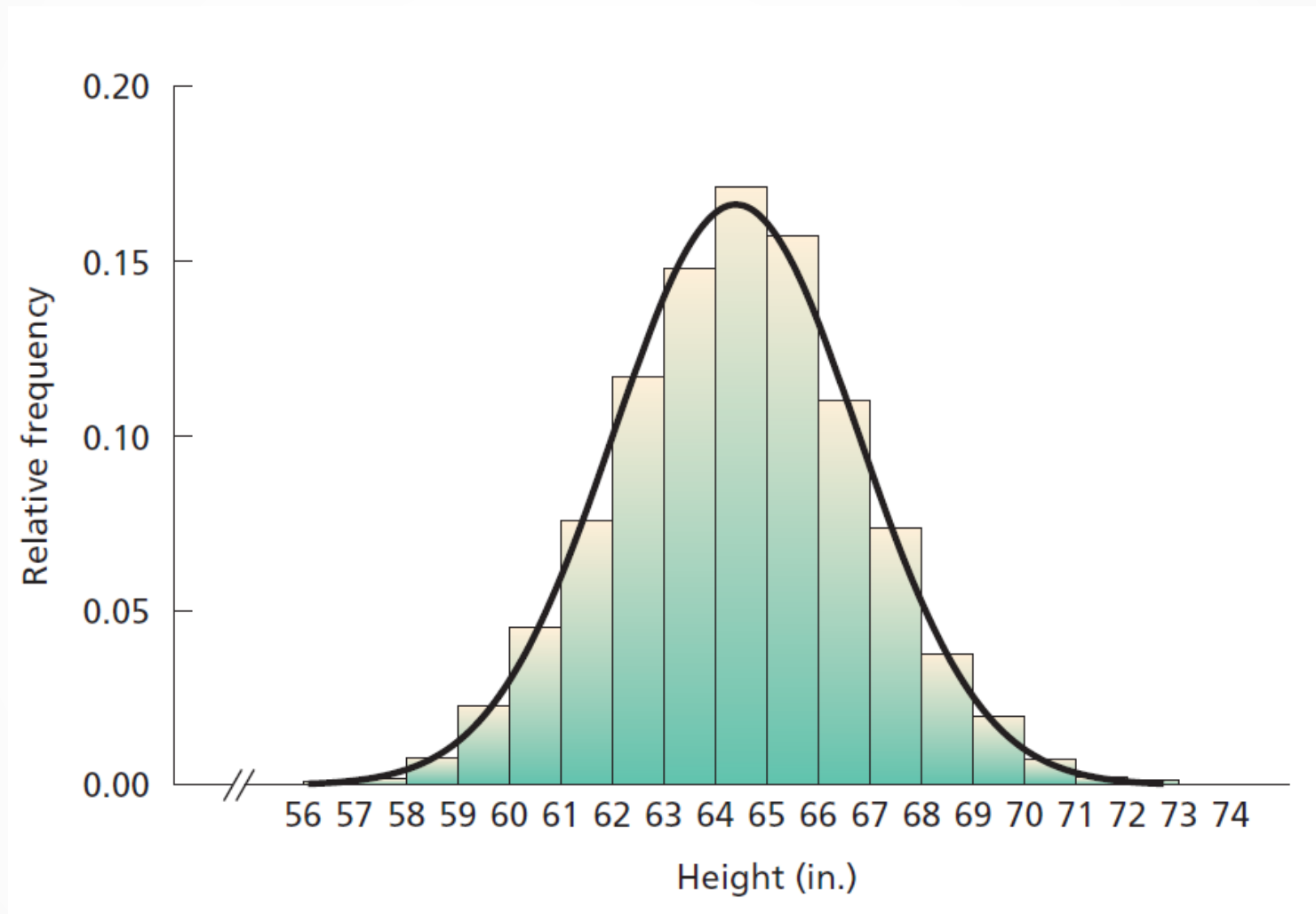
frequency distributions, relative-frequency distributions, frequency histograms, relative-frequency histograms, dotplots, stem-and-leaf diagrams, pie charts, and bar charts.

Distributive Shape

An important aspect of the distribution of a quantitative data set is its shape. In-deed, as we demonstrate in later chapters, the shape of a distribution frequently plays a role in determining the appropriate method of statistical analysis. To identify the shape of a distribution, the best approach usually is to use a smooth curve that approximates the overall shape.

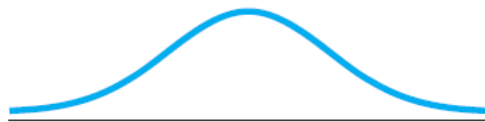
Distribution of a Data Set

Distribution Shape: Example



Distribution of a Data Set

Distribution Shapes



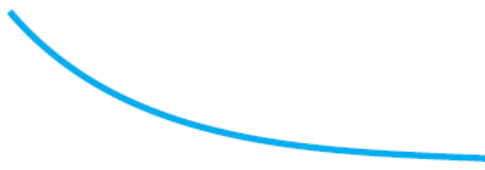
(a) Bell shaped



(b) Triangular



(c) Uniform (or rectangular)



(d) Reverse J shaped



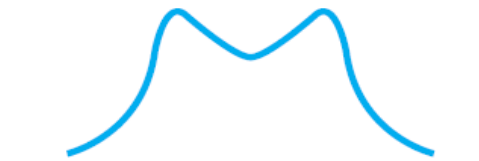
(e) J shaped



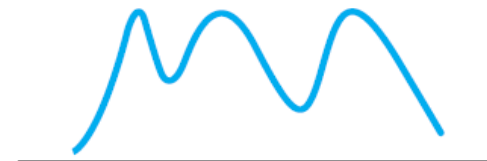
(f) Right skewed



(g) Left skewed



(h) Bimodal

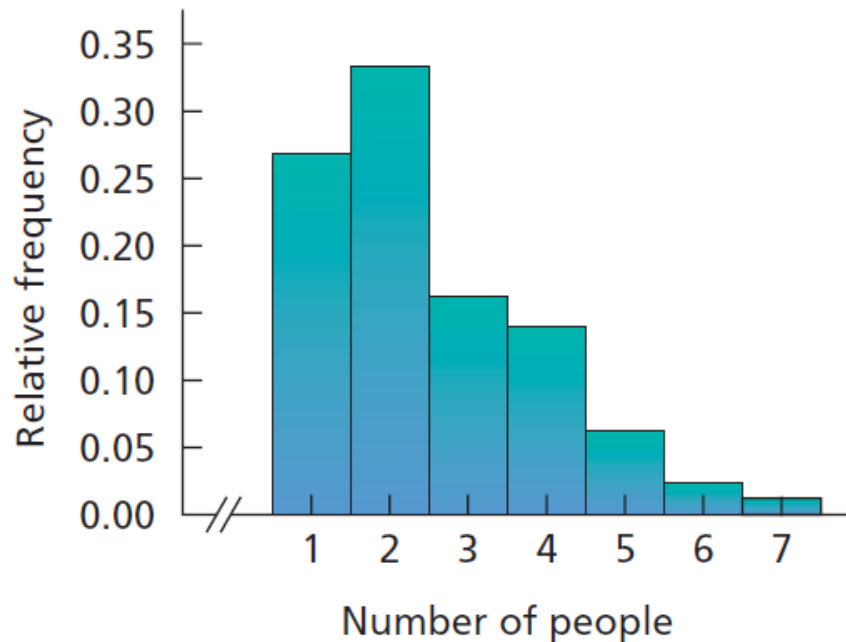


(i) Multimodal

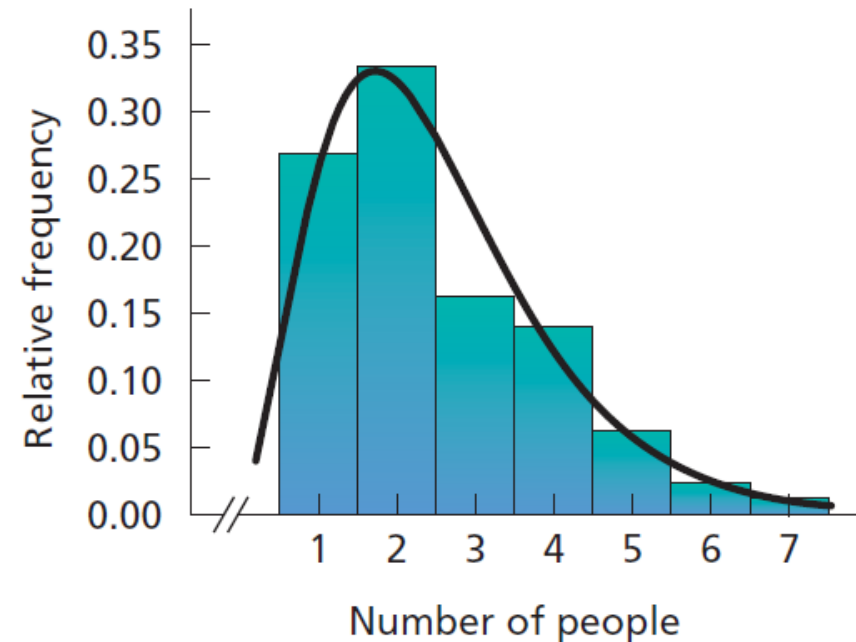
Distribution of a Data Set

Identifying Distribution Shapes

Household Size The relative-frequency histogram for household size in the United States shown in Figure is based on data contained in Current Population Reports, a publication of the U.S. Census Bureau. Identify the distribution shape for sizes of U.S. households.



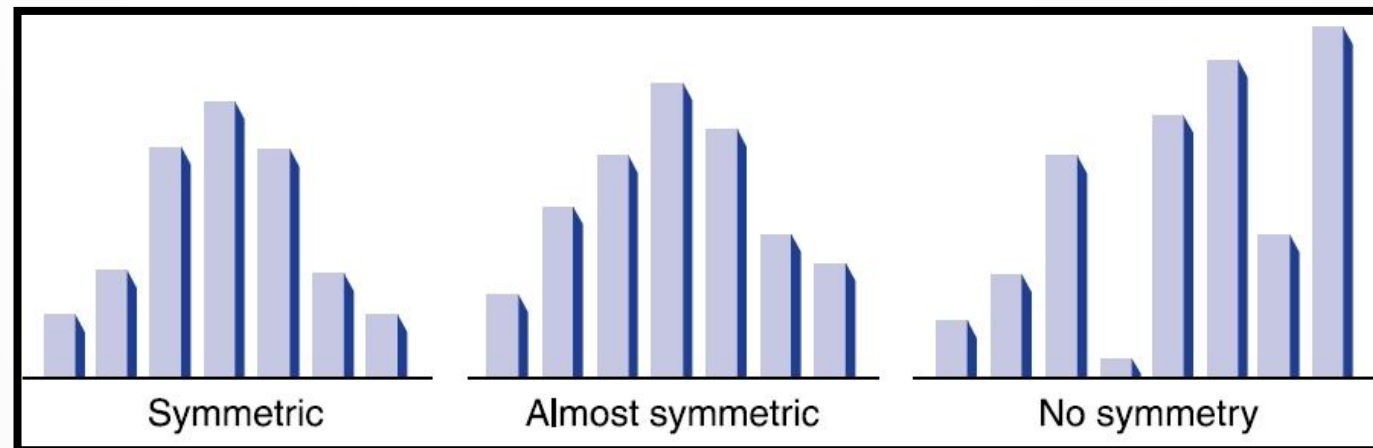
(a)



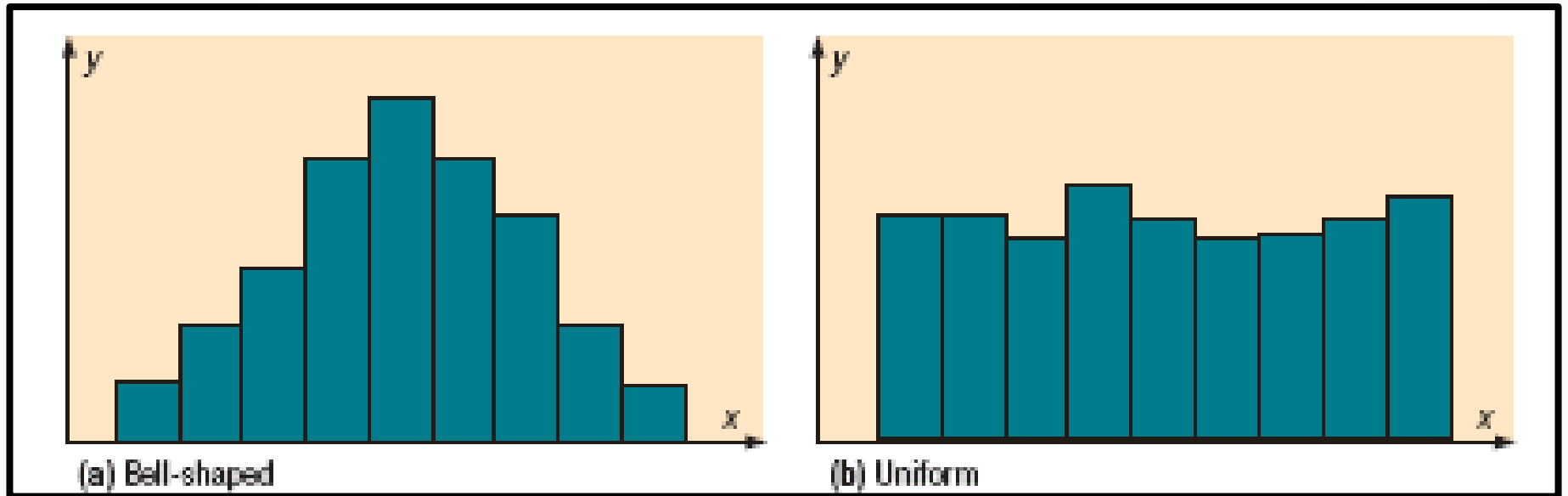
(b)

Approximately Symmetric

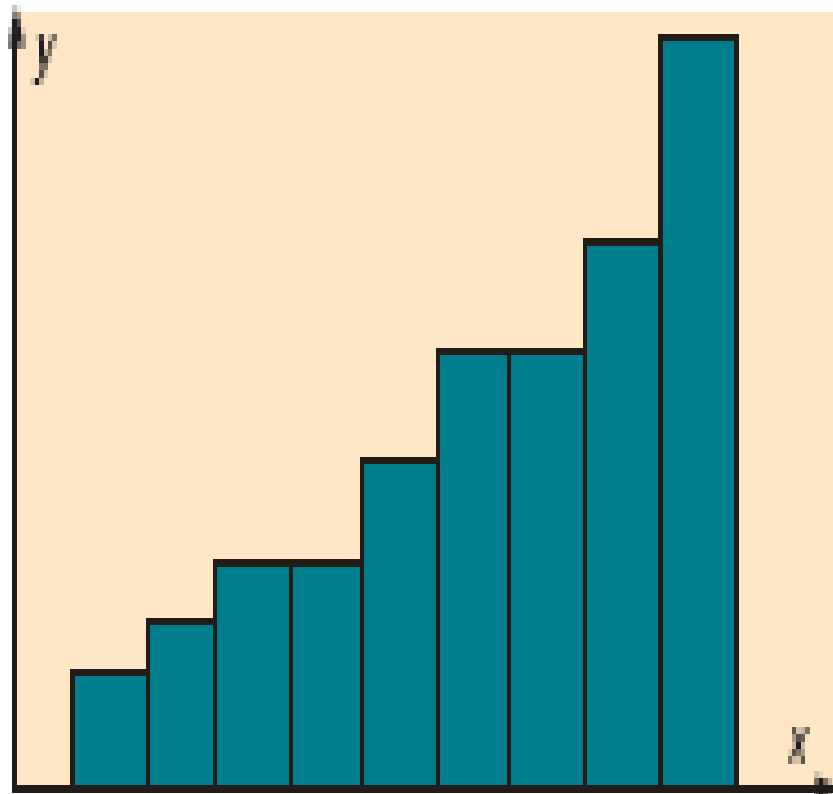
- Data that are “close to” being symmetric are said to be *approximately symmetric*.
- one of a symmetric data set,
- one of an approximately symmetric data set, and
- one of a data set that exhibits no symmetry.



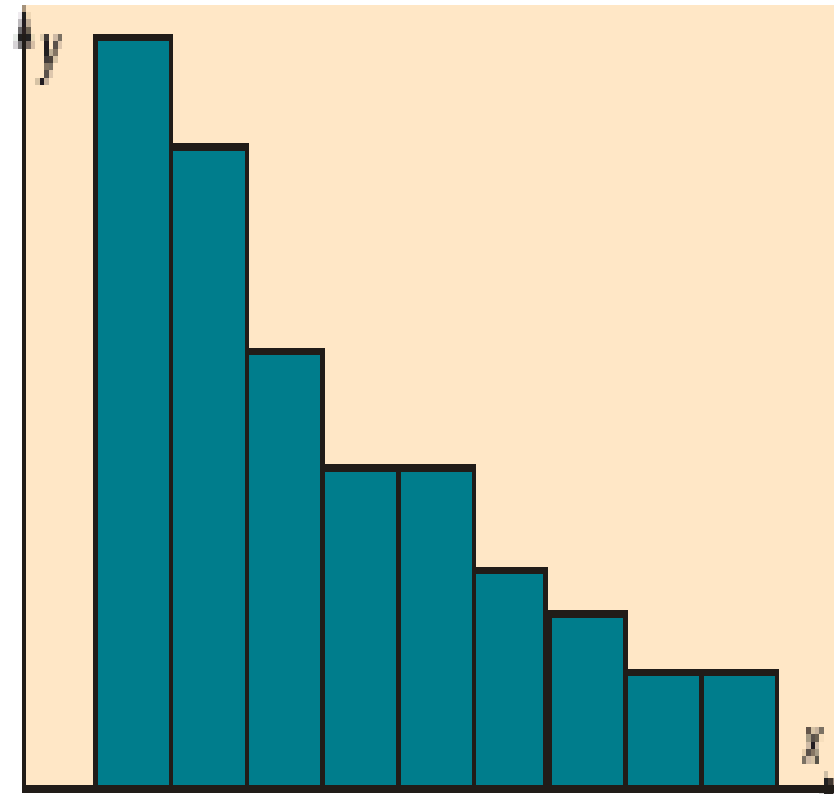
Quick Review of histogram shapes



Cont...

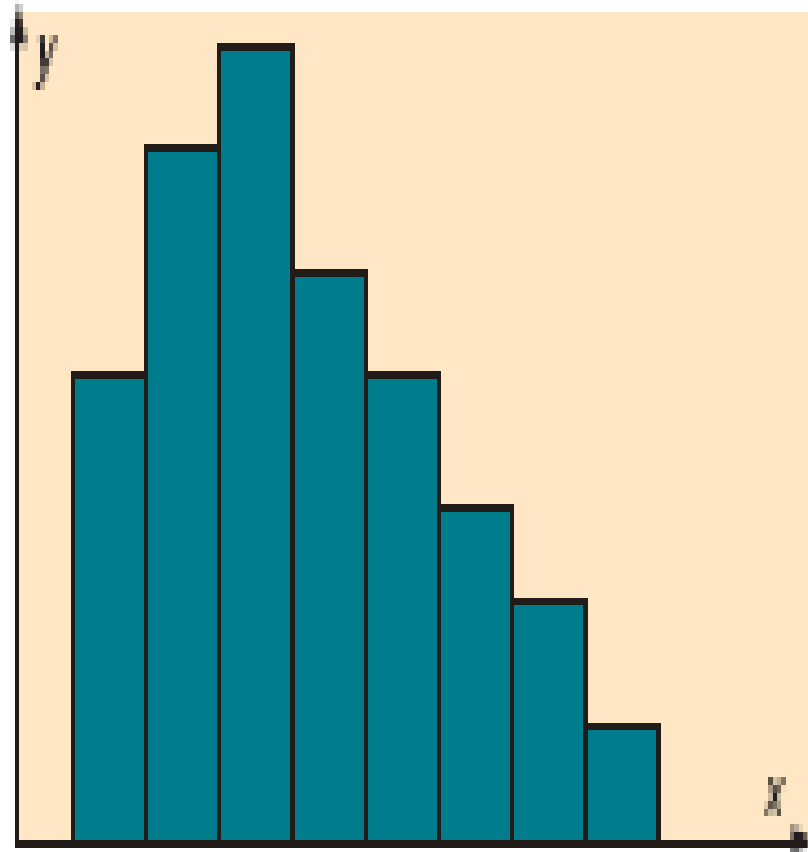


(c) J-shaped

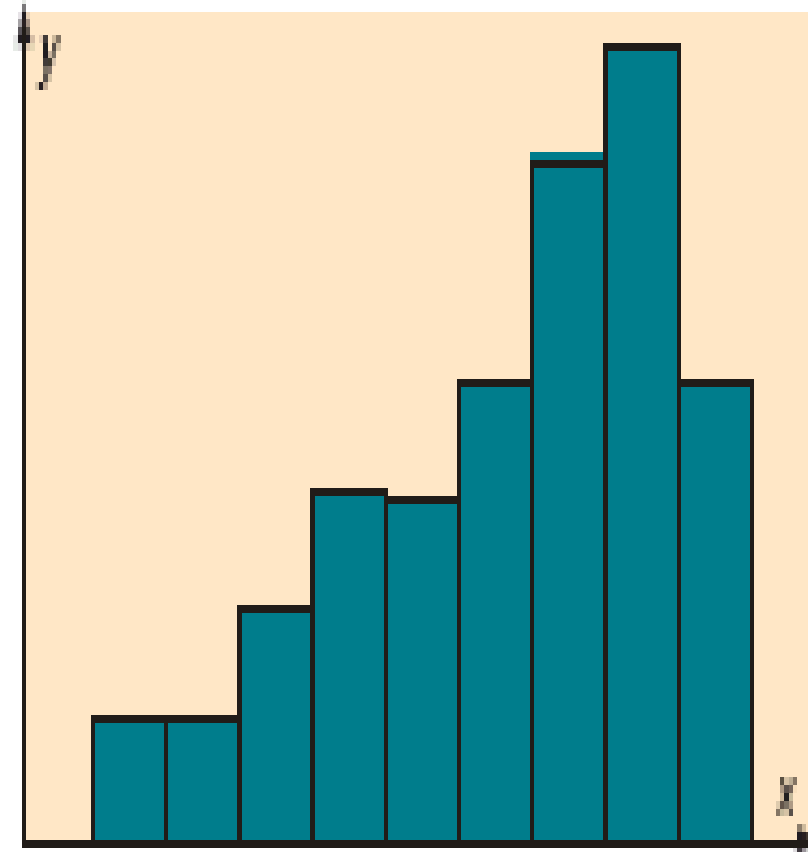


(d) Reverse J-shaped

Cont...



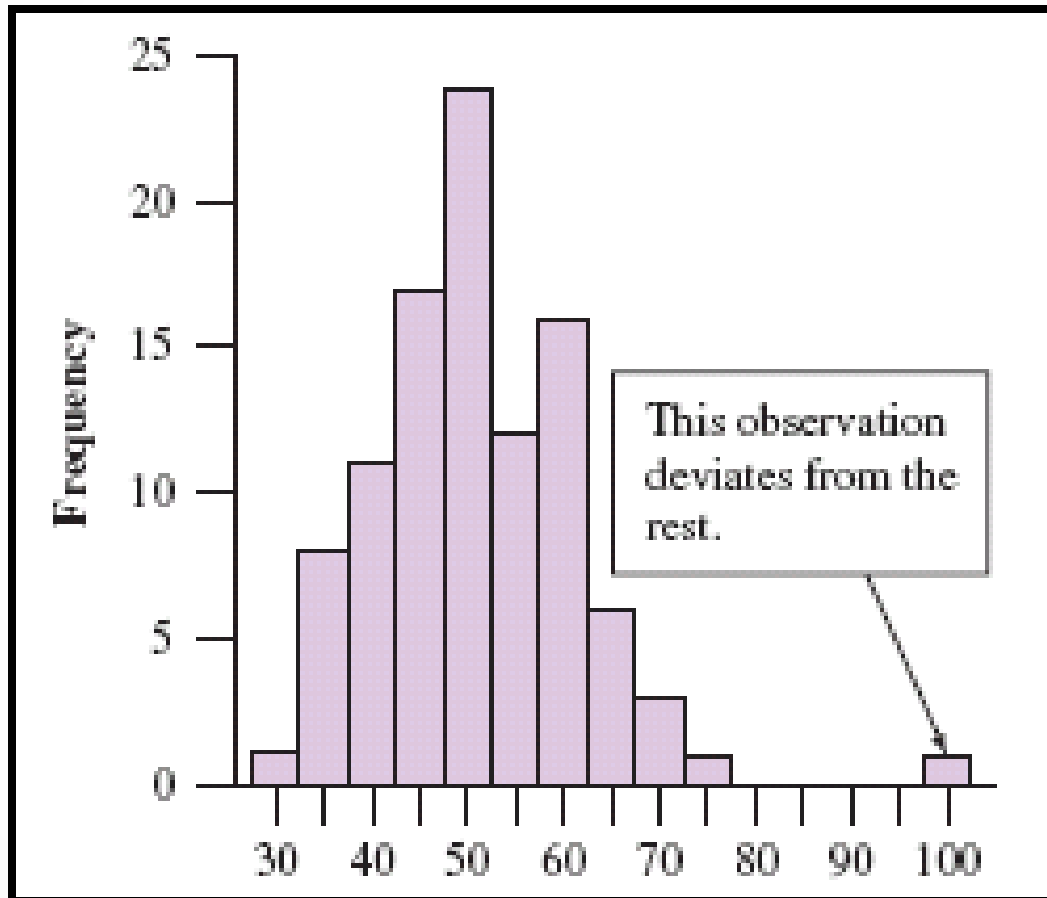
(e) Right-skewed



(f) Left-skewed

Outlier

An **outlier** falls far from the rest of the data





Ogive frequency

Cumulative Frequency Curve or Ogive

Cumulative frequency of a particular frequency (in a given frequency distribution) is calculated by adding all the frequencies above to that particular frequency.

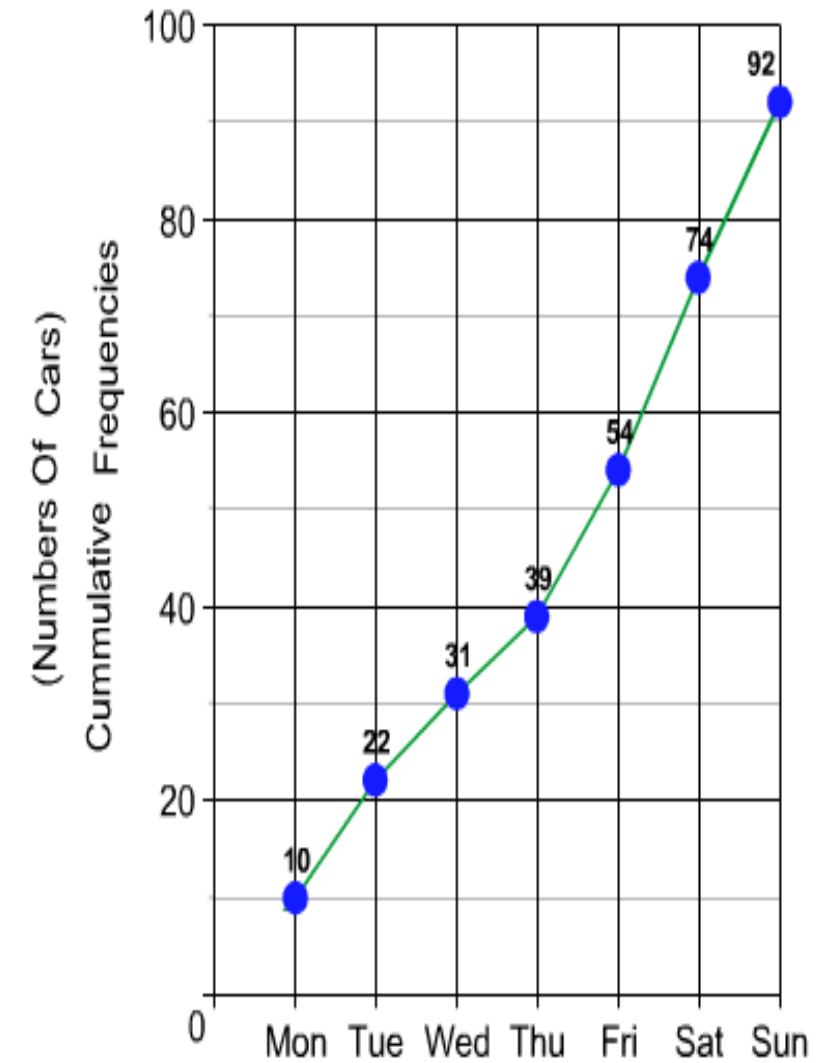
Cumulative frequency curve is a curve which is drawn by plotting cumulative frequency points on a graph.

Cumulative histograms, also known as ogives, are graphs that can be used to determine **how many data values lie above or below a particular value in a data set.**

Example: Following table shows how cumulative frequency of a given data is computed:

Cont...

Days	Number of cars parked (f)	Cumulative Frequencies (c.f.)
Monday	10	10
Tuesday	12	$10 + 12 = 22$
Wednesday	9	$22 + 9 = 31$
Thursday	8	$31 + 8 = 39$
Friday	15	$39 + 15 = 54$
Saturday	20	$54 + 20 = 74$
Sunday	18	$74 + 18 = 92$
	Total = 92	



Types of Ogive

- ❑ Less than ogive
- ❑ More than ogive

Less than ogive method

To construct a cumulative frequency polygon and an ogive by less than method, we use the following algorithm.

Algorithm

Step 1 : Start with the upper limits of class intervals and add class frequencies to obtain the cumulative frequency distribution.

Step 2 : Mark upper class limits along X-axis on a suitable scale.

Step 3 : Mark cumulative frequencies along Y-axis on a suitable scale.

Step 4 : Plot the points (x_i, f_i) where x_i is the upper limit of a class and f_i is corresponding cumulative frequency.

Step 5 : Join the points obtained in step 4 by a free hand smooth curve to get the ogive and to get the cumulative frequency polygon join the points obtained in step 4 by line segments.

Example

The following table shows the daily sales of 230 footpath sellers of Chandni Chowk.

Sales in Rs.	No. of sellers
0 – 500	12
500 – 1000	18
1000 – 1500	35
1500 – 2000	42
2000 – 2500	50
2500 – 3000	45
3000 – 3500	20
3500 – 4000	8

Cont...

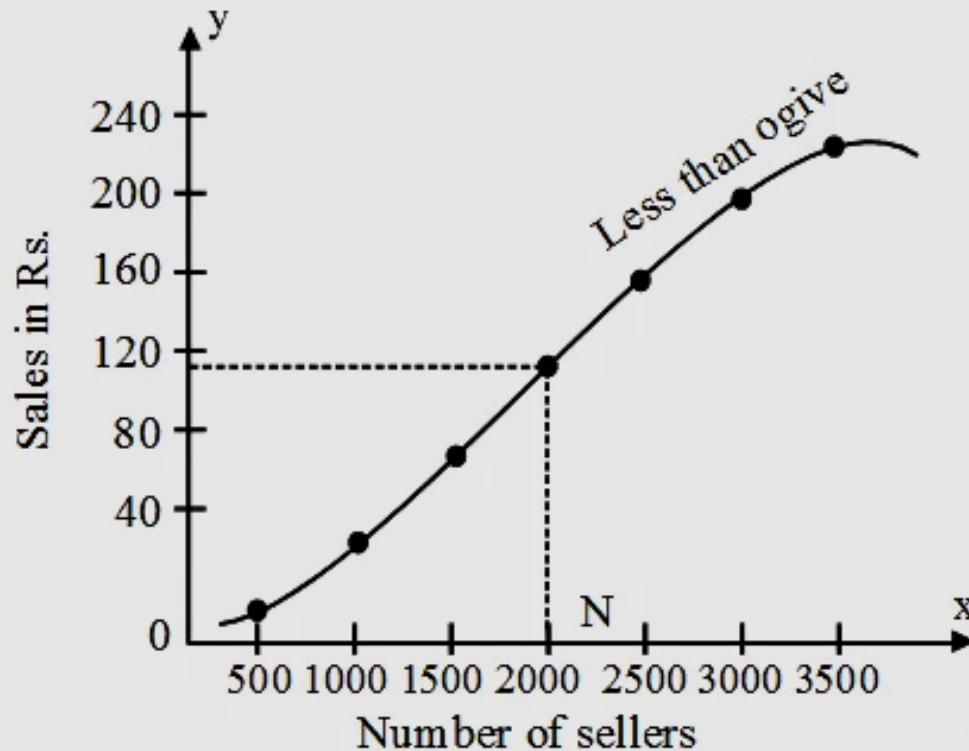
Solution: To draw ogive, we need to have a cumulative frequency distribution.

Sales in Rs.	No. of sellers	Less than type cumulative frequency
0 – 500	12	12
500 – 1000	18	30
1000 – 1500	35	65
1500 – 2000	42	107
2000 – 2500	50	157
2500 – 3000	45	202
3000 – 3500	20	222
3500 – 4000	8	230

Cont...

Less than ogive :

Sales in Rs. are taken on the y-axis and number of sellers are taken on x-axis. For drawing less than ogive, points (500, 12), (1000, 30), (1500, 65), (2000, 107), (2500, 157), (3000, 202), (3500, 222), (4000, 230) are plotted on graph paper and these are joined free hand to obtain the less than ogive.



N = total frequency

The value $\frac{N}{2} = 115$ is marked on y-axis and a line parallel to x-axis is drawn. This line meets the curve at a point P. From P draw a perpendicular PN to meet x-axis at median. Median = 2000. Hence, the median of given frequency distribution is 2000.

More than ogive method

To construct a cumulative frequency polygon and an ogive by more than method, we use the following algorithm.

Algorithm

Step 1 : Start with the lower limits of the class intervals and from the total frequency subtract the frequency of each class to obtain the cumulative frequency distribution.

Step 2 : Mark the lower class limits along X-axis on a suitable scale.

Step 3 : Mark the cumulative frequencies along Y-axis on a suitable scale.

Step 4 : Plot the points (x_i, f_i) where x_i is the lower limit of a class and f_i is corresponding cumulative frequency.

Step 5 : Join the points obtained in step 4 by a free hand smooth curve to get the ogive and to get the cumulative frequency polygon join these points by line segments

Example

Example 3: Draw the two ogives for the following frequency distribution of the weekly wages of (less than and more than) number of workers.

Weekly wages	Number of workers
0 – 20	41
20 – 40	51
40 – 60	64
60 – 80	38
80 – 100	7

Cont...

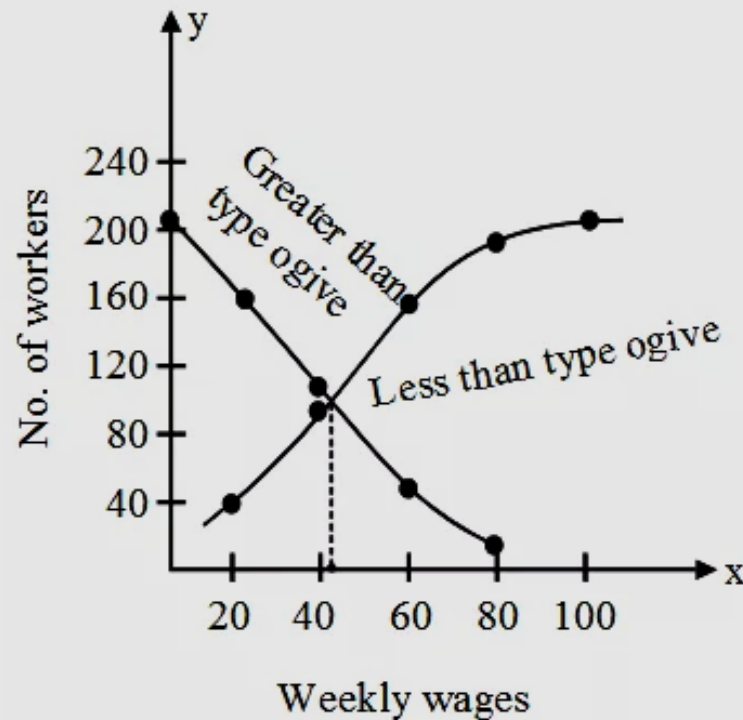
Solution:

Weekly wages	Number of workers	C.F (less than)	C.F (More than)
0 – 20	41	41	201
20 – 40	51	92	160
40 – 60	64	156	109
60 – 80	38	194	45
80 – 100	7	201	7

Cont...

Less than curve :

Upper limits of class intervals are marked on the x-axis and less than type cumulative frequencies are taken on y-axis. For drawing less than type curve, points (20, 41), (40, 92), (60, 156), (80, 194), (100, 201) are plotted on the graph paper and these are joined by free hand to obtain the less than ogive.



Cont...

Greater than ogive

Lower limits of class interval are marked on x-axis and greater than type cumulative frequencies are taken on y-axis. For drawing greater than type curve, points (0, 201), (20, 160), (40, 109), (60, 45) and (80, 7) are plotted on the graph paper and these are joined by free hand to obtain the greater than type ogive. From the point of intersection of these curves a perpendicular line on x-axis is drawn. The point at which this line meets x-axis determines the median. Here the median is 42.652.

Example 1:

The table given below shows the marks obtained by 80 students in science. Construct (i) less than ogive (ii) more than ogive.

Markes	No. of students
0-10	3
10-20	8
20-30	17
30-40	29
40-50	15
50-60	6
60-70	2

Solution:

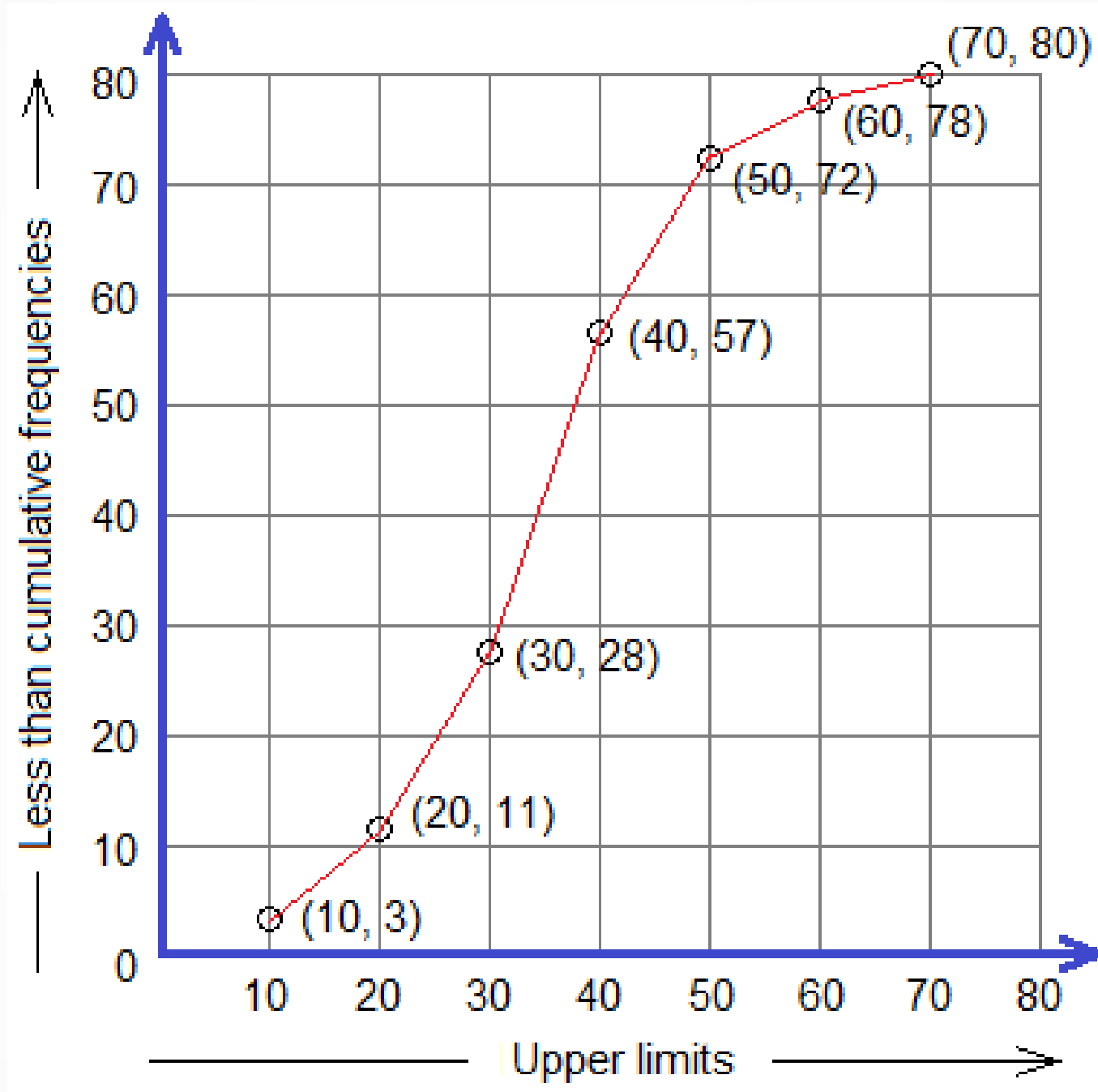
Here,

Less than cumulative frequency distribution table:

Marks	No. of students	Upper limit	Less than c.f.
0-10	3	10	3 (less than 10)
10-20	8	20	11 (less than 20)
20-30	17	30	28 (less than 30)
30-40	29	40	57 (less than 40)
40-50	15	50	72 (less than 50)
50-60	6	60	78 (less than 60)
60-70	2	70	80 (less than 70)

Here, we have the coordinates to draw less than ogive: (10, 3), (20, 11), (30, 28), (40, 57), (50, 72), (60, 78) and (70, 80).

Plotting these points on a graph, we have the following less than ogive



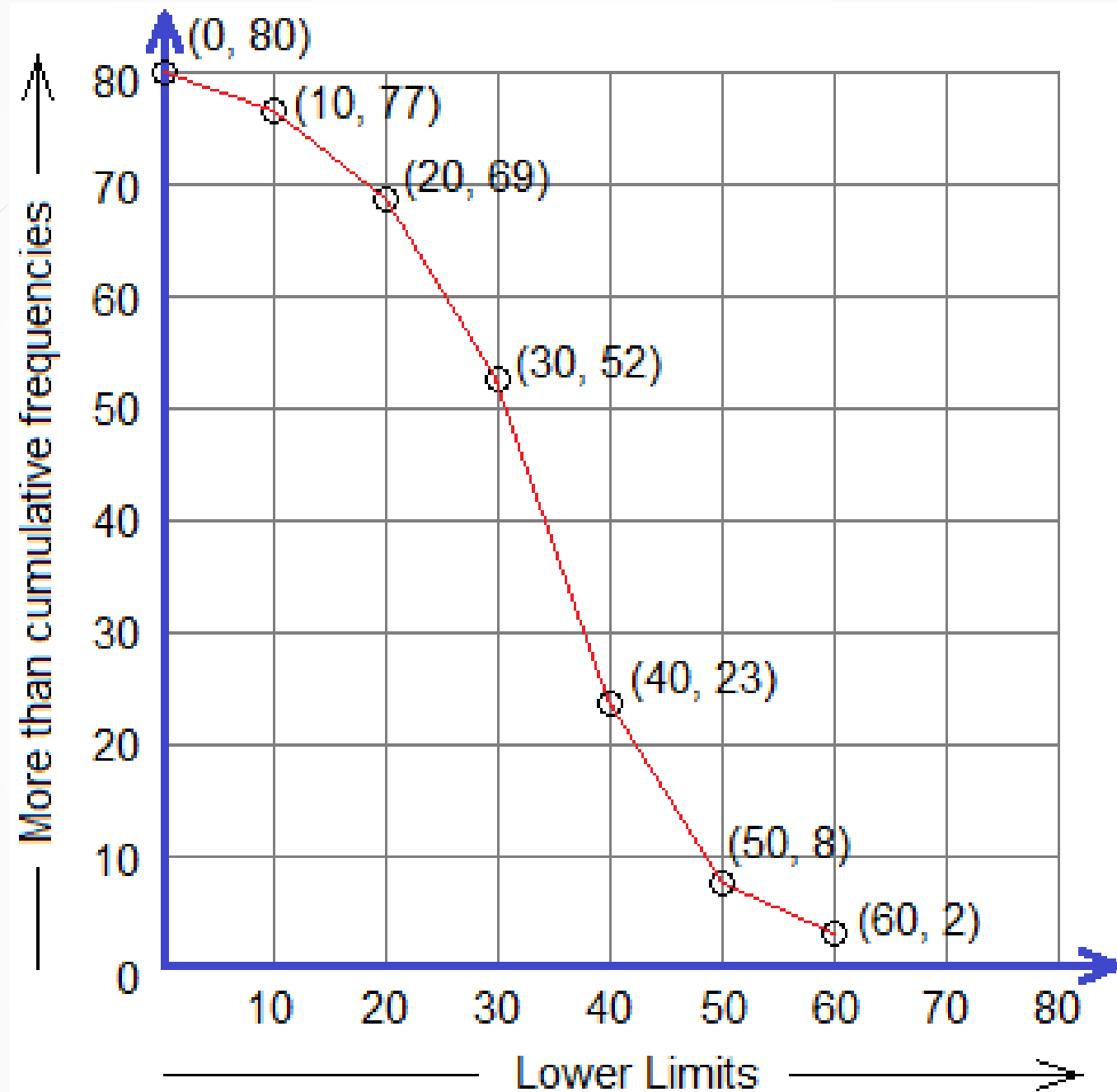


More than cumulative frequency distribution table:

Marks	No. of students	Upper limit	More than c.f.
0-10	3	0	80 (more than 0)
10-20	8	10	77 (more than 10)
20-30	17	20	69 (more than 20)
30-40	29	30	52 (more than 30)
40-50	15	40	23 (more than 40)
50-60	6	50	8 (more than 50)
60-70	2	60	2 (more than 60)

Here, we have the coordinates to draw more than ogive: (0, 80), (10, 77), (20, 69), (30, 52), (40, 23), (50, 8) and (60, 2).

Plotting these points on a graph, we have the following more than ogive.





Less than and more than (Combined) Ogive

If we draw both less than and more than ogive of a distribution on the same graph paper, it is called **combined ogive**. The point of intersection of two ogives in a combined ogive is called the **vital-point**.

The foot of the perpendicular drawn from vital-point to x-axis gives the value of the median of the distribution. And, the corresponding class is the proper class for the median.

Uses of ogive

Following are the uses of cumulative frequency curve (ogive):

- 1.** Ogive or cumulative frequency curve helps us to determine as well as portray the number or proportion of cases above or below the given value

Example 2: Construct the combined ogive from the data given in the table of Example 1, and find the median.

Solution: Here,

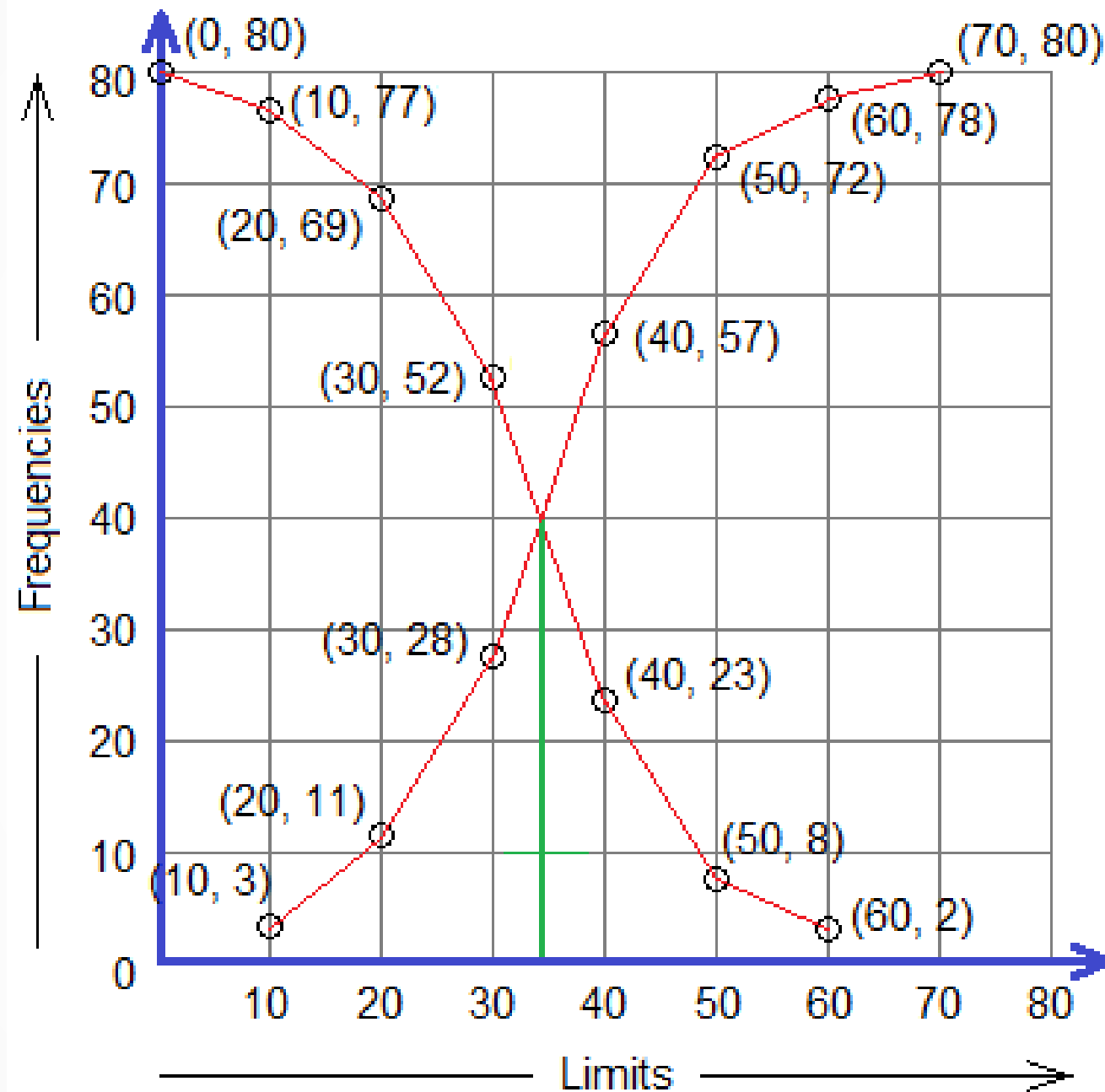
Less than and more than cumulative frequency distribution table:

Marks	f	Upper limit	Less than c.f.	Lower limit	More than c.f.
0-10	3	10	3	0	80
10-20	8	20	11	10	77
20-30	17	30	28	20	69
30-40	29	40	57	30	52
40-50	15	50	72	40	23
50-60	6	60	78	50	8
60-70	2	70	80	60	2

Here, we have the coordinates to draw less than ogive: (10, 3), (20, 11), (30, 28), (40, 57), (50, 72), (60, 78) and (70, 80).

Here, we have the coordinates to draw more than ogive: (0, 80), (10, 77), (20, 69), (30, 52), (40, 23), (50, 8) and (60, 2).

Plotting these points on a graph, we have the following less than and more than combined ogive.





DOT PLOT

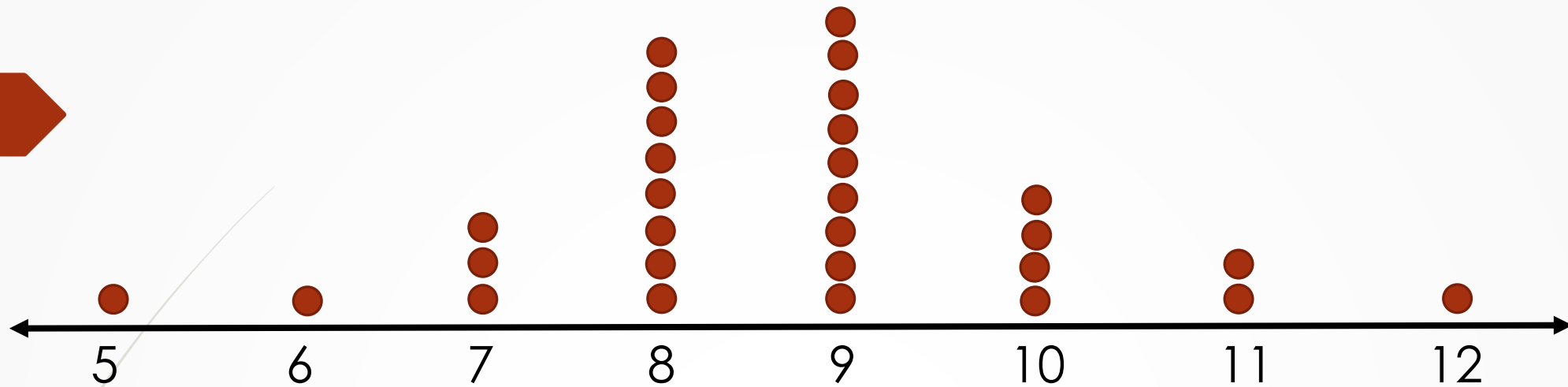


7 8 5 9 9 9 7 7 10 10 11 9 8 8 8 12 6
11 10 8 8 9 9 9 8 10 9 9 8

- Casey decided to make a dot plot of the data to help him answer his statistical question. He first drew a number line and labeled it from 5 to 12 to match the lowest and highest number of hours slept.
- He then placed a dot above 7 for the first piece of data he collected. He continued to place dots above each number until each number was represented by

**Please finish
making
Casey's dot
plot.**





1. What are the **least** and **most** hours of sleep reported in the survey of 8th graders? **5 and 12 hours of sleep.**
2. What is the modal number of hours slept? **9 hours of sleep.**
(Modal means most common value or most frequently occurring value.)
3. How many hours of sleep describes the center of data? **8 to 9 hours of sleep.**

Dot Plots

- A dot plot is made up of dots plotted on a graph.
- Each dot represents a specific number of observations from a set of data.
- The dots are stacked in a column over a category, so that the height of the column represents the relative or absolute frequency of observations in the category.
- The pattern of data in a dot plot can be described in terms of **symmetry** and **skewness** only if the categories are quantitative.
- Dot plots are used most often to **plot frequency counts** within a small number of categories, usually with small sets of data.
- Dot plots are great ways to allow us to identify **the spread** of the data and **the mode** of the data.

Dot Plots – How to draw...

1. Determine the highest and lowest values.
2. Draw a number line that starts at the lowest and finishes at the highest.
3. Now place a dot above the number for the first data entry and then a dot above the next number for the second data entry and so on.
4. If you get to a value that already has a dot then put another dot above this one.
5. The dots need to be evenly spaced to give an accurate picture.

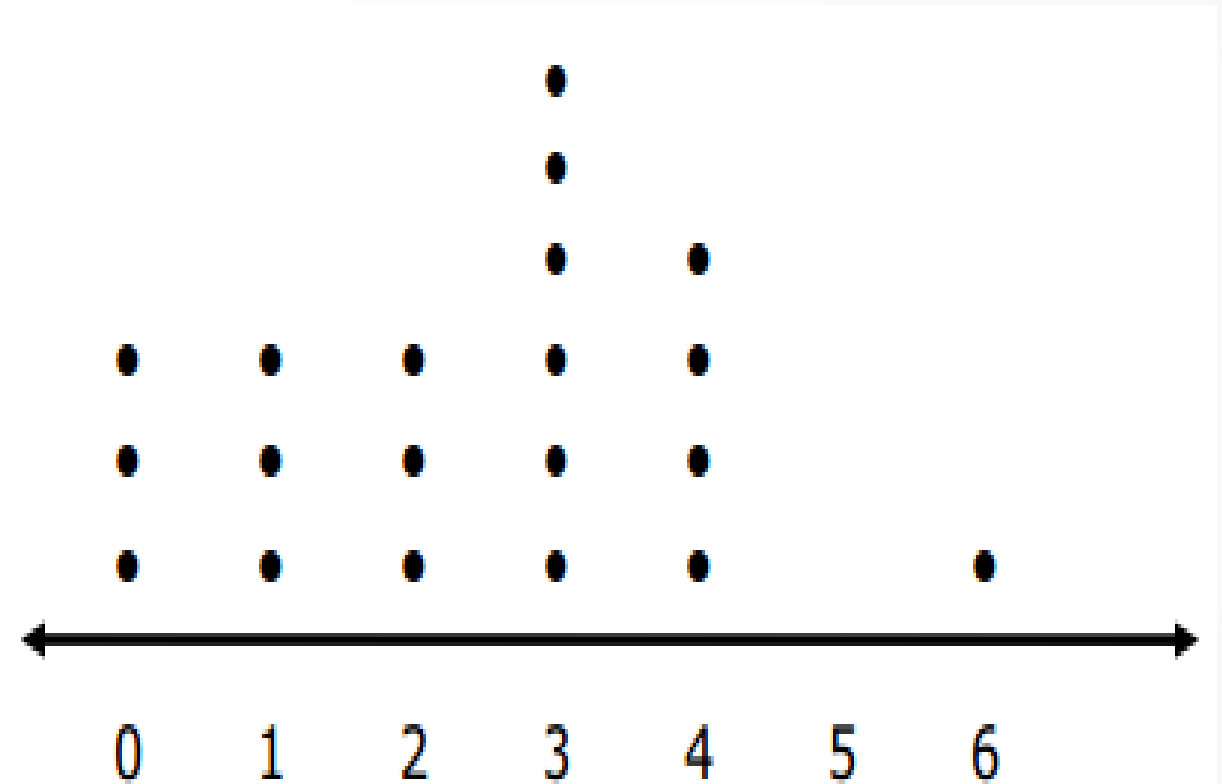
Create a Dot Plot

- The students in Mr. Furman's social studies 1st period class were asked how many brothers and sisters (siblings) they each have.

- Here are the results:

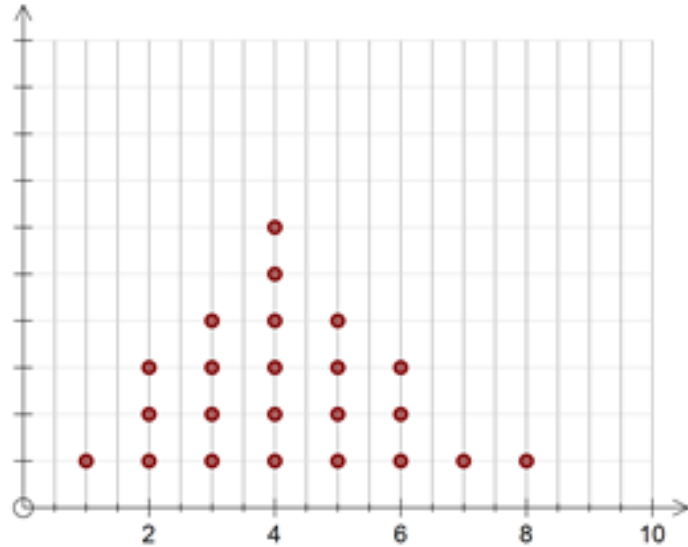
➤ 4 0 3 3 0 4 4 0 1 6 1 3 1
2 3 2 3 4 2 3

- Draw a box plot of this data.



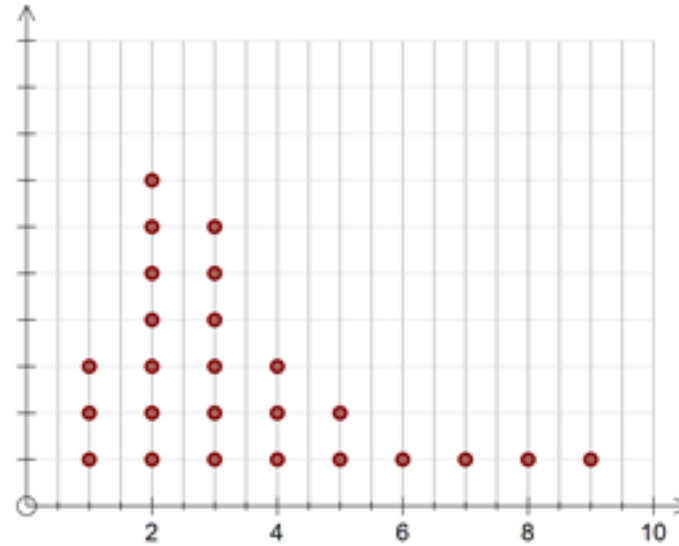
Describing the Spread of Dot Plots

Symmetrical



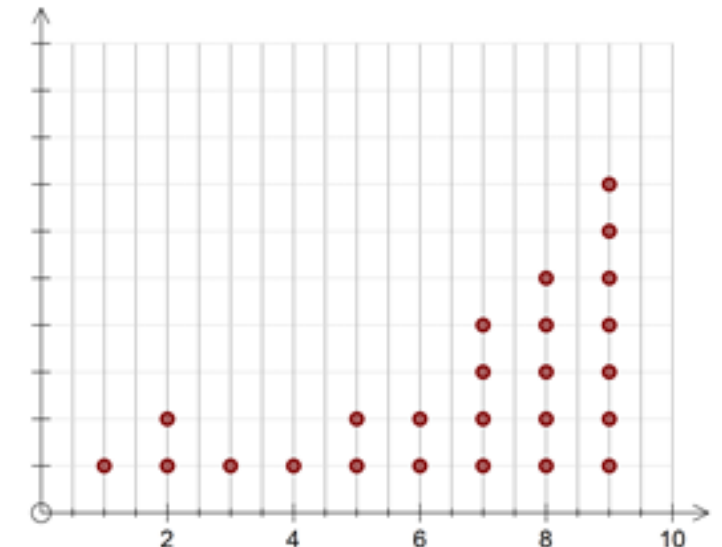
If you connected the top dot of each column, it would form a symmetrical curve.

Skewed Right



There are few observations on the right, so the data is skewed right.

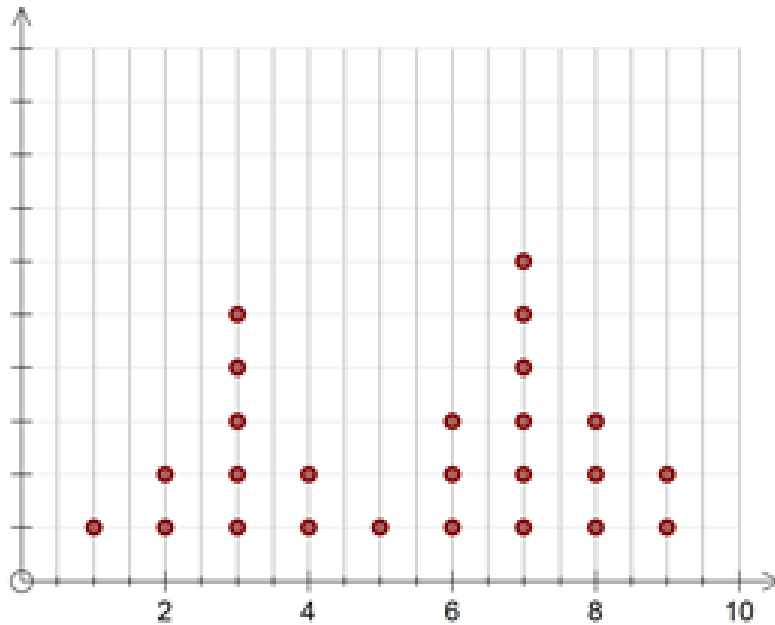
Skewed Left



There are fewer observations on the left, so the data is skewed left.

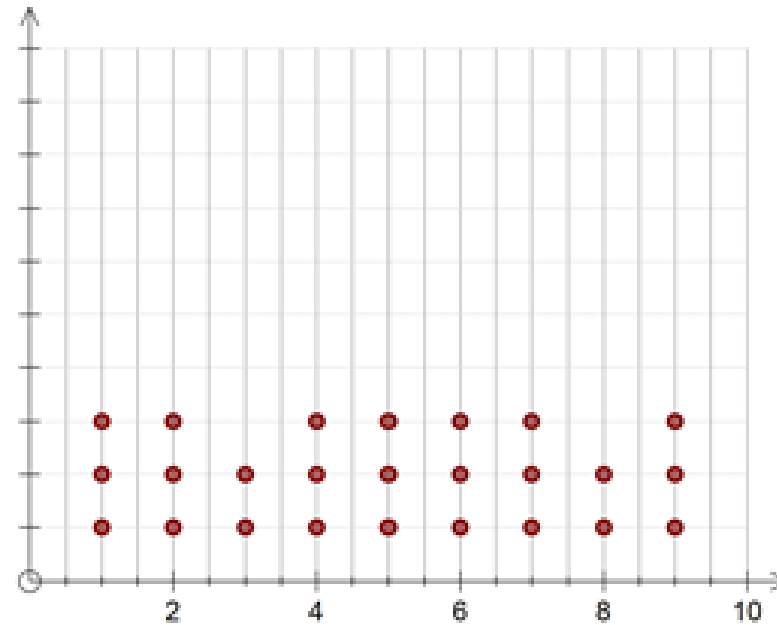
Describing the Spread of Dot Plots

Bi modal

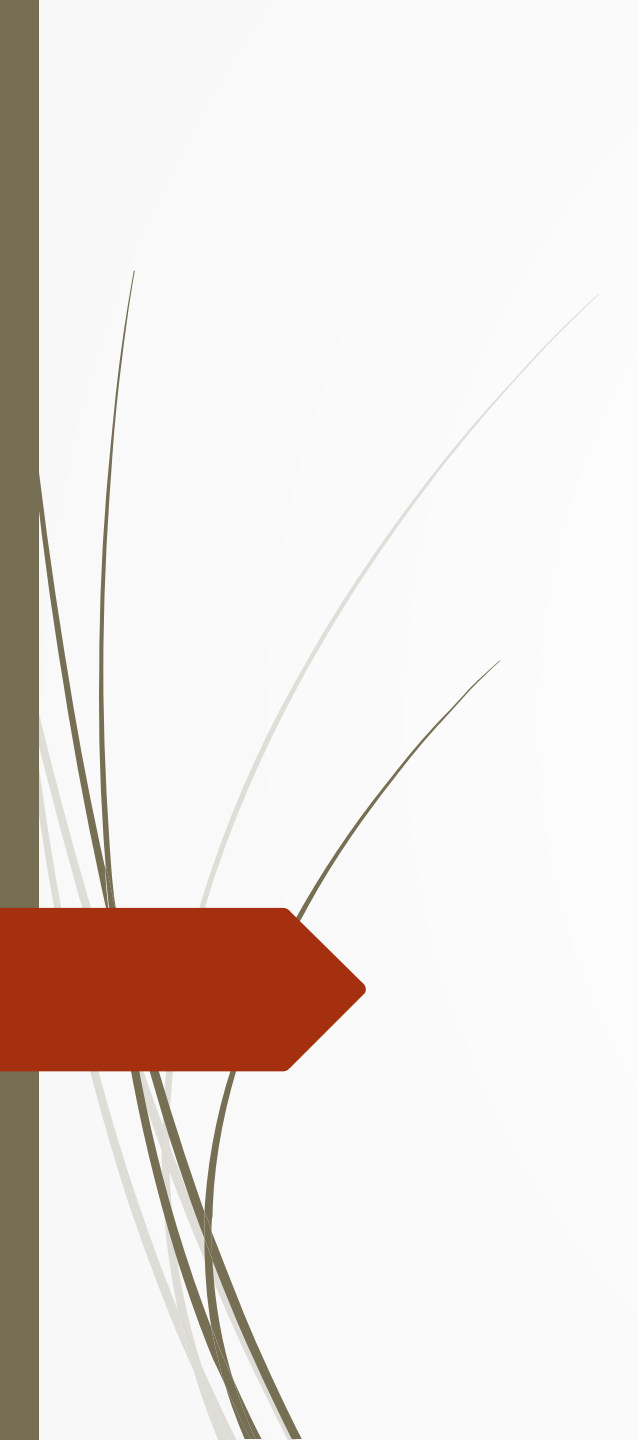


The data has two areas where it peaks.

Uniform



The data is about the same for all numbers.



Stem-and-Leaf Plots

Susan Phillips
Lee's Summit, MO



Stem-and-Leaf Plot

- * Uses place value to organize data
- * Shows data in an organized way so it can be analyzed easily
- * Organizes data so it is easier to find the median, mode, and range

Here are the scores for a freshman basketball team

Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41

Points Earned				
Stem	Leaves			
1	2	6	8	
2	0	7	7	9
3	2	3	4	
4	1	2	8	
5	1			

Key: 1 | 2 = 12

- * Here is the same data organized into a stem-and-leaf plot.

Can you tell how the stem-and-leaf plot was made?

Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41

The first number in the data is 27.

Points Earned

Stem	Leaves
1	2 6 8
2	0 7 7 9
3	2 3 4
4	1 2 8
5	1

Here is the 2 in the tens place.

Here is the 7 in the ones place.

Key: 1 | 2 = 12

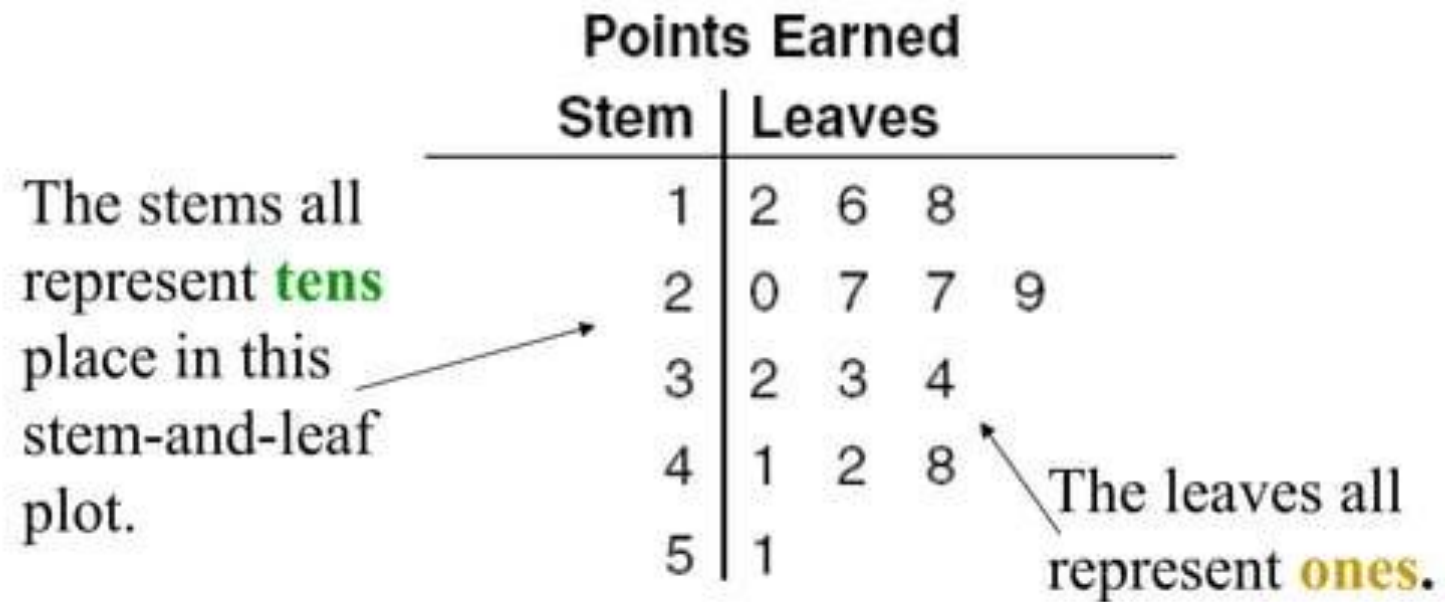
Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41

Points Earned				
Stem	Leaves			
1	2	6	8	
2	0	7	7	9
3	2	3	4	
4	1	2	8	
5	1			

Key: 1 | 2 = 12

The key shows us which place value the digits represent.

Points Earned in Games During Basketball Season						
27	16	34	29	48	12	33
20	18	42	51	27	32	41



Key: 1 | 2 = 12



**Let's
try one
together...**



Here are one student's math test scores.
Make a stem-and-leaf plot for this data.

Test Scores				
75	86	83	91	94
88	84	99	79	86

First, draw
the dividing
lines.



Here are one student's math test scores.
Make a stem-and-leaf plot for this data.

Test Scores				
75	86	83	91	94
88	84	99	79	86

Stems	Leaves

Next, label
the stem side
and the leaf
side.

Here are one student's math test scores.
Make a stem-and-leaf plot for this data.

Test Scores				
75	86	83	91	94
88	84	99	79	86

Stems	Leaves
7	5

Find the
smallest
piece of data
and plot it on
the grid.

Here are one student's math test scores.
Make a stem-and-leaf plot for this data.

Test Scores				
75	86	83	91	94
88	84	99	79	86

Stems	Leaves
7	5 9

Find the next
smallest
piece of data
and plot it on
the grid.

The 7 for the tens
place is already
there. We just
add the 9.

Continue with this process until
you have entered all the data.

Test Scores				
75	86	83	91	94
88	84	99	79	86

Stems	Leaves
7	5 9
8	3 4 6 6 8
9	1 4 9

Remember to add a title.

Test Scores				
75	86	83	91	94
88	84	99	79	86

Test Scores

Stems	Leaves
7	5 9
8	3 4 6 6 8
9	1 4 9

Don't forget the key!

Test Scores				
75	86	83	91	94
88	84	99	79	86

Test Scores

Stems	Leaves
7	5 9
8	3 4 6 6 8
9	1 4 9

Key: 7 | 5 means 75

**You just
completed a
stem-and-leaf plot!**



**Now, try it
on your own...**



This table shows the times in seconds and hundredths of seconds for the women's 500-meter speed-skating competition in the 2002 Winter Olympics.

74.75	74.94	75.19	75.37	75.39	75.64	75.64	76.17	76.20	76.31
76.37	76.42	76.62	76.73	76.73	76.86	76.92	77.10	77.37	77.60
77.60	77.71	78.26	78.63	78.79	78.89	79.28	79.45	79.45	

Use this data to make a stem-and-leaf plot.

2002 Winter Olympics 500-Meter
Speed-Skating Times in Seconds

74.75	74.94	75.19	75.37	75.39	75.64	75.64	76.17	76.20	76.31
76.37	76.42	76.62	76.73	76.73	76.86	76.92	77.10	77.37	77.60
77.60	77.71	78.26	78.63	78.79	78.89	79.28	79.45	79.45	

Stems	Leaves
74	75 94
75	
76	
77	
78	
79	

Key: 74 | 75 means 74.75

Check your work...

2002 Winter Olympics 500-Meter
Speed-Skating Times in Seconds

Stems	Leaves
74	75 94
75	19 37 39 64 64
76	17 20 31 37 42 62 73 73 86 92
77	10 37 60 60 71
78	26 63 79 89
79	28 45 45

Key: 74 | 75 means 74.75

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220

Stems	Leaves
4	0 0 1 5 7
5	1 1 2 4
6	3 3 3 5 9 9
7	0 4 4
8	3 6 7
9	1 4

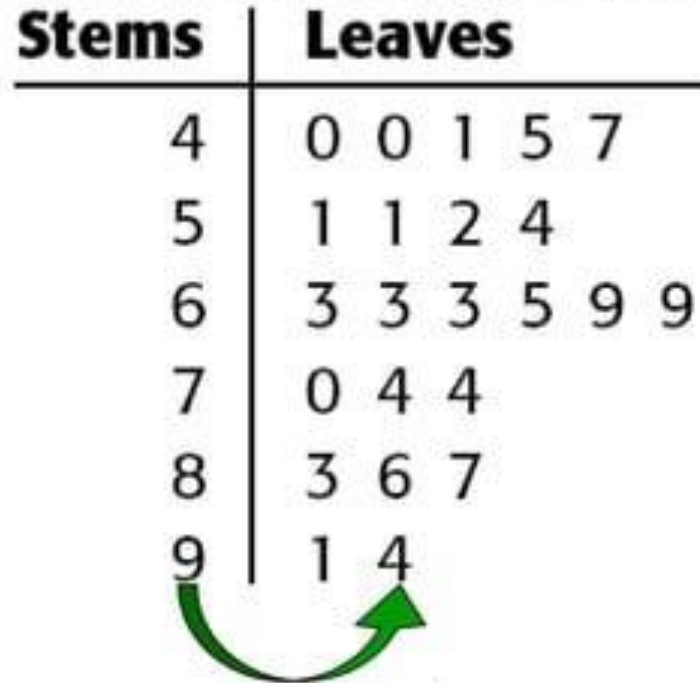
Key: 4|0 means 40

- * What was the least number of cookies sold?

40 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220



Key: 4|0 means 40

- * What was the largest number of cookies sold?

94 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220

Stems	Leaves
4	0 0 1 5 7
5	1 1 2 4
6	3 3 3 5 9 9
7	0 4 4
8	3 6 7
9	1 4

Key: 4|0 means 40

* Find the median of the data

Find the middle leaf...

63 boxes

Using Stem-and-Leaf Plots

Boxes of Girl Scout Cookies Sold by Troop 220

Stems	Leaves
4	0 0 1 5 7
5	1 1 2 4
6	3 3 3 5 9 9
7	0 4 4
8	3 6 7
9	1 4

Key: 4|0 means 40

- * Find the mode of the data

Look for the number that occurs
most often in a row of leaves

63 boxes

Stem-and-Leaf Plots

The
End



